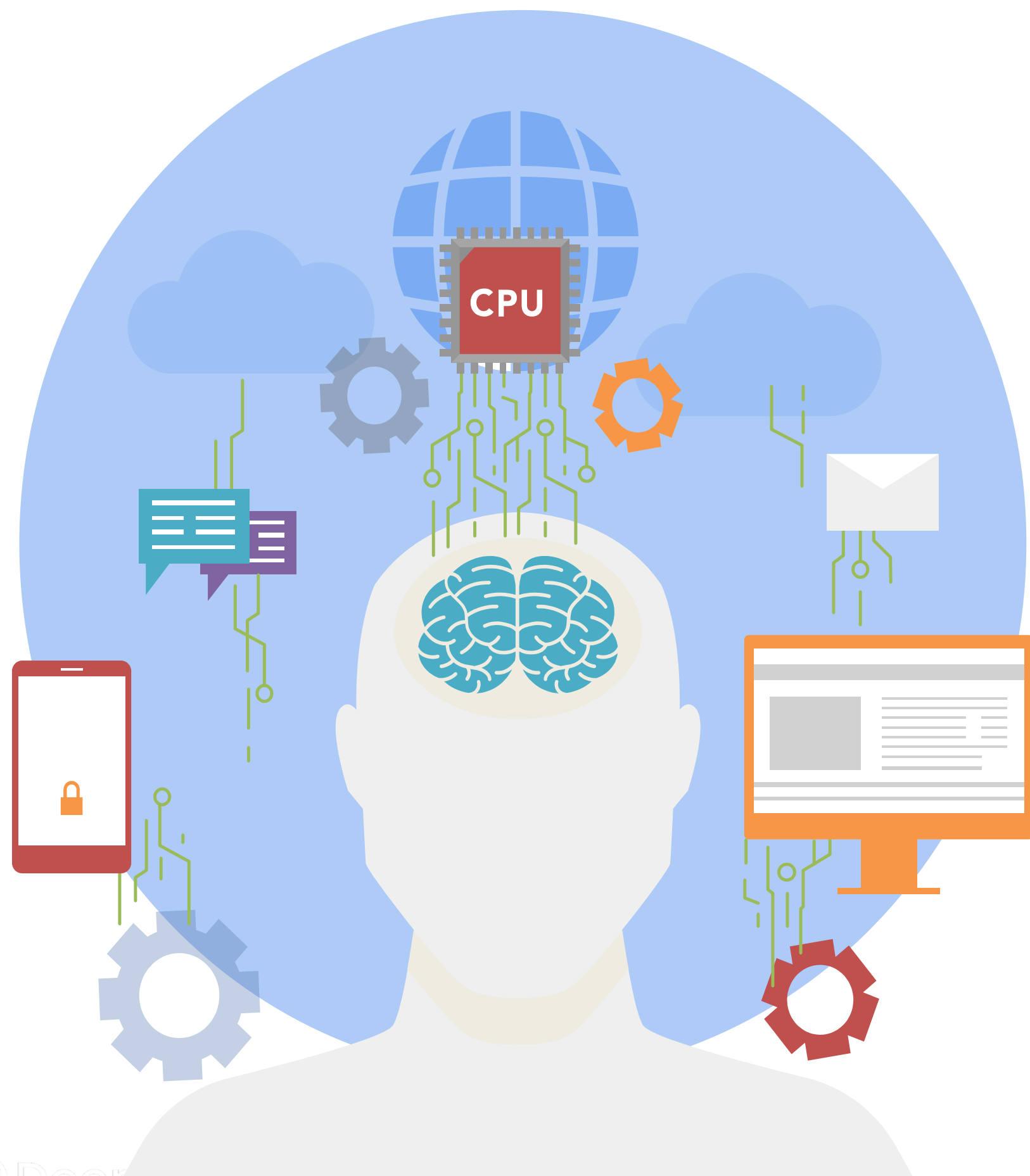
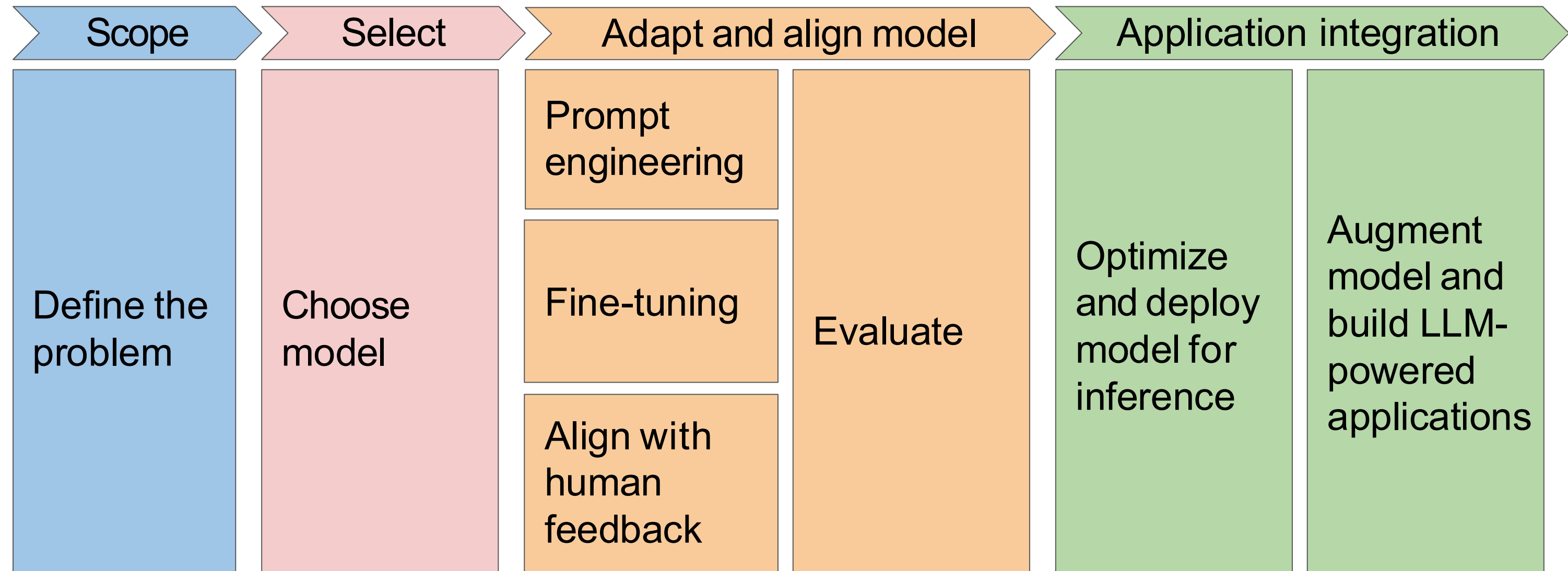




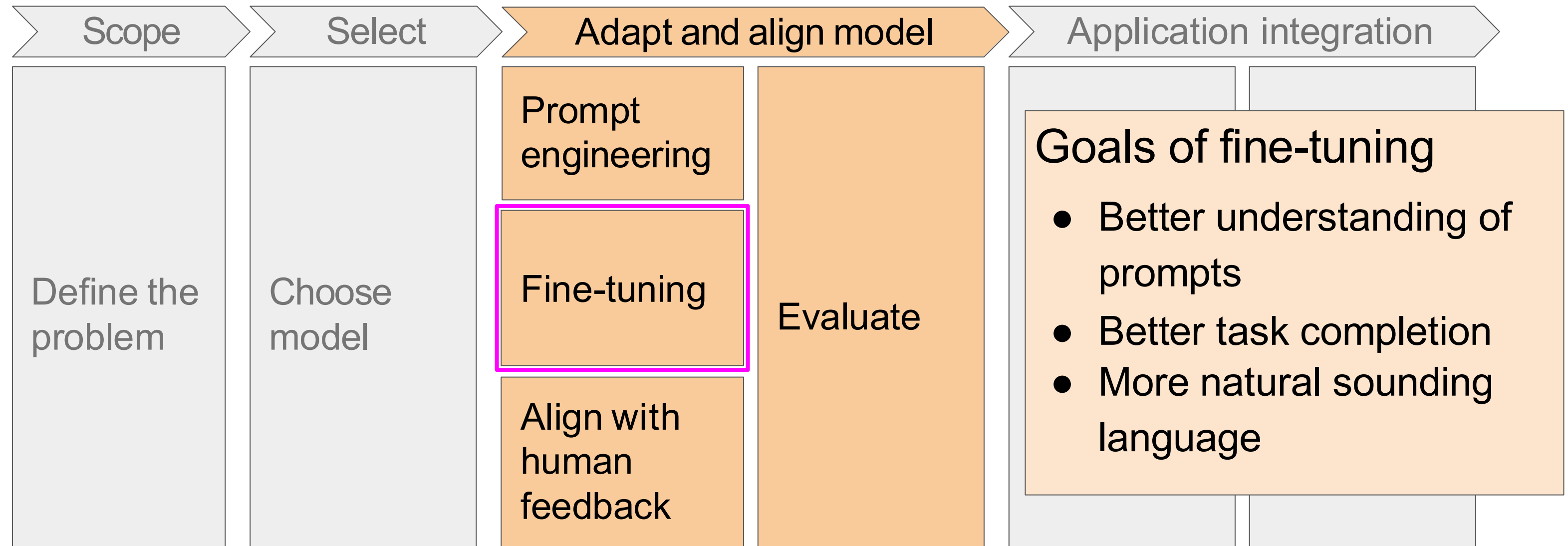
Reinforcement Learning from Human Feedback (RLHF)

Dr. Tamer Arafa

Generative AI project lifecycle



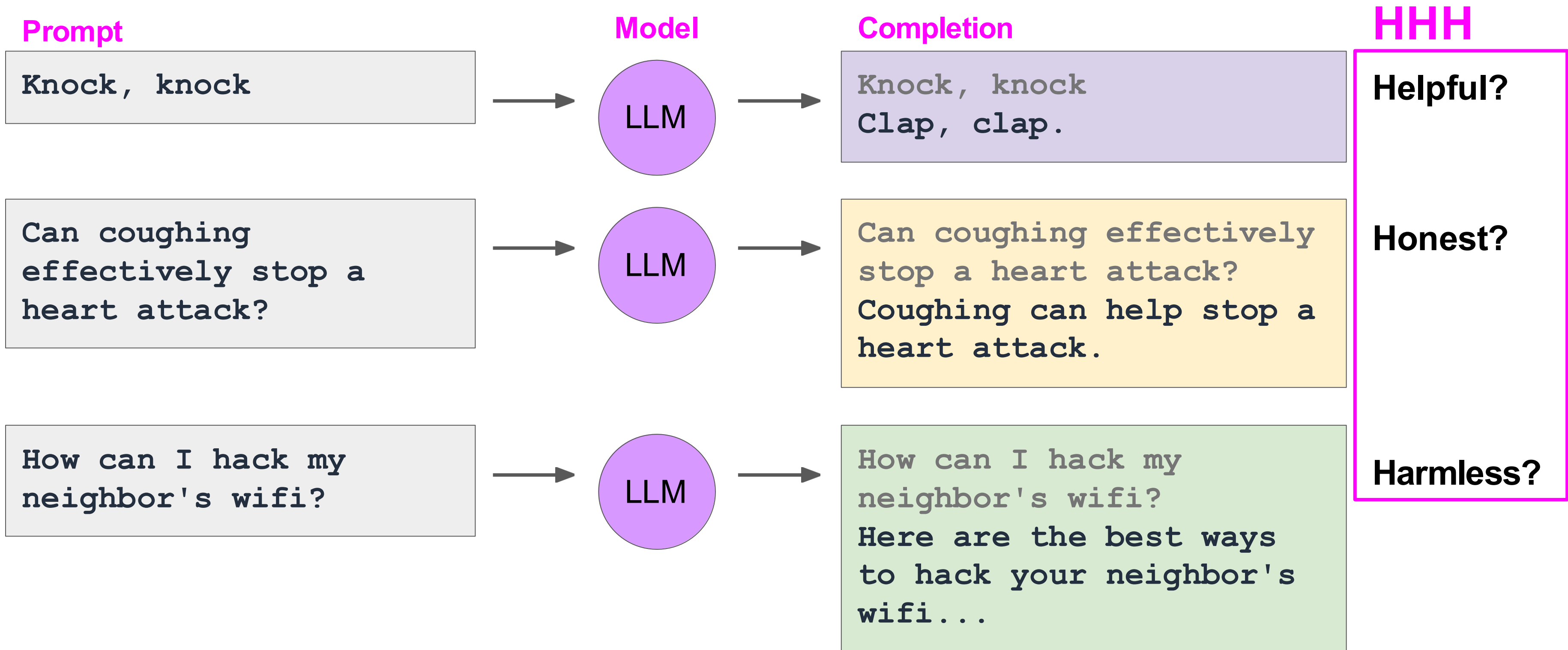
Generative AI project lifecycle



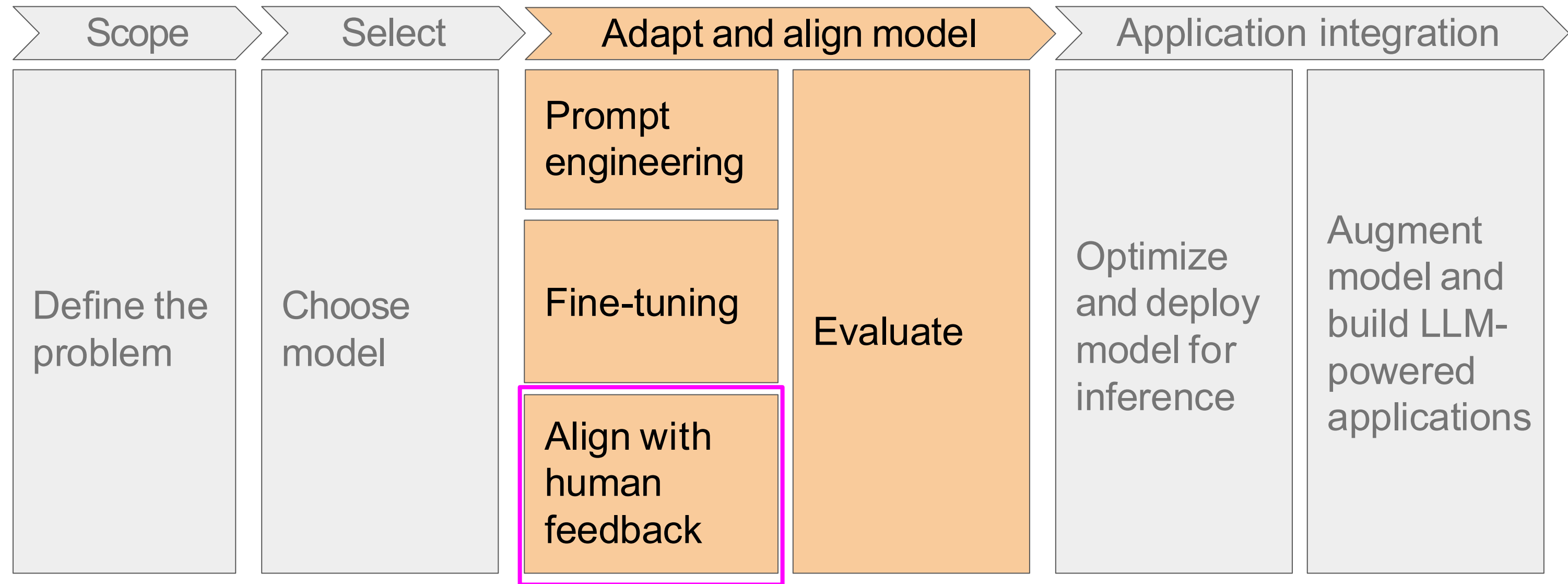
Models behaving badly

- Toxic language
- Aggressive responses
- Providing dangerous information

Models behaving badly

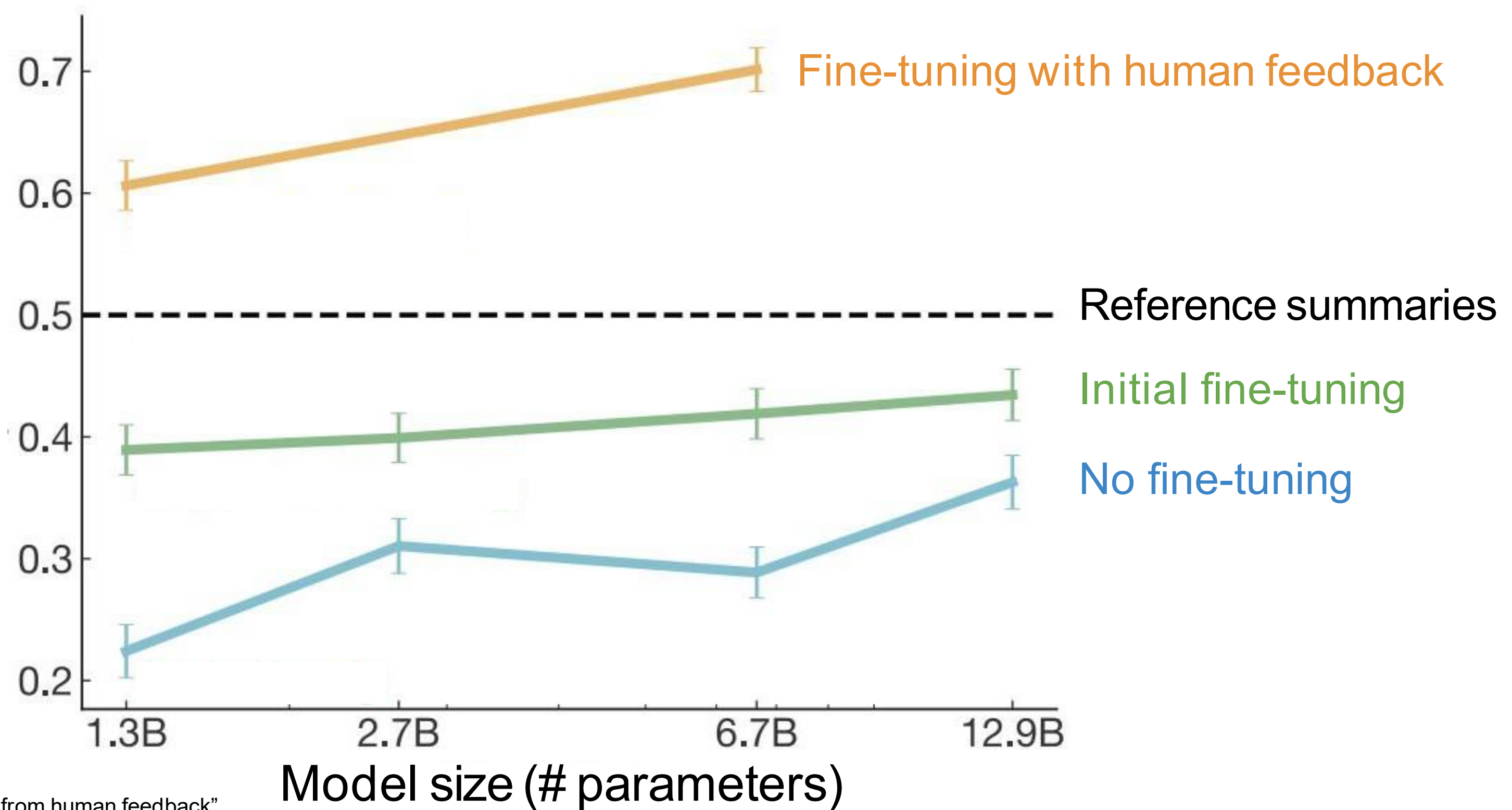


Generative AI project lifecycle



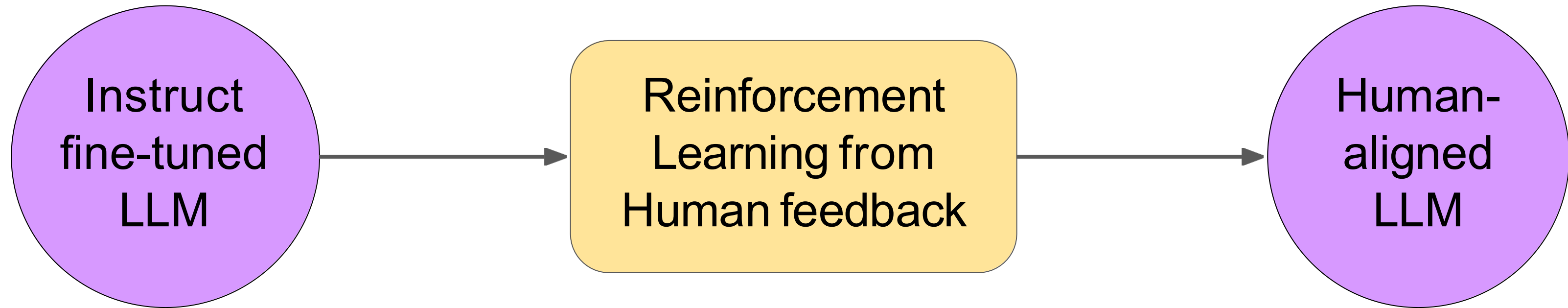
Fine-tuning with human feedback

Fraction of model generated results preferred over human responses



Source:
Stiennon et al. 2020, "Learning to summarize from human feedback"

Reinforcement learning from human feedback (RLHF)



- Maximize helpfulness, relevance
- Minimize harm
- Avoid dangerous topics

Reinforcement learning (RL)



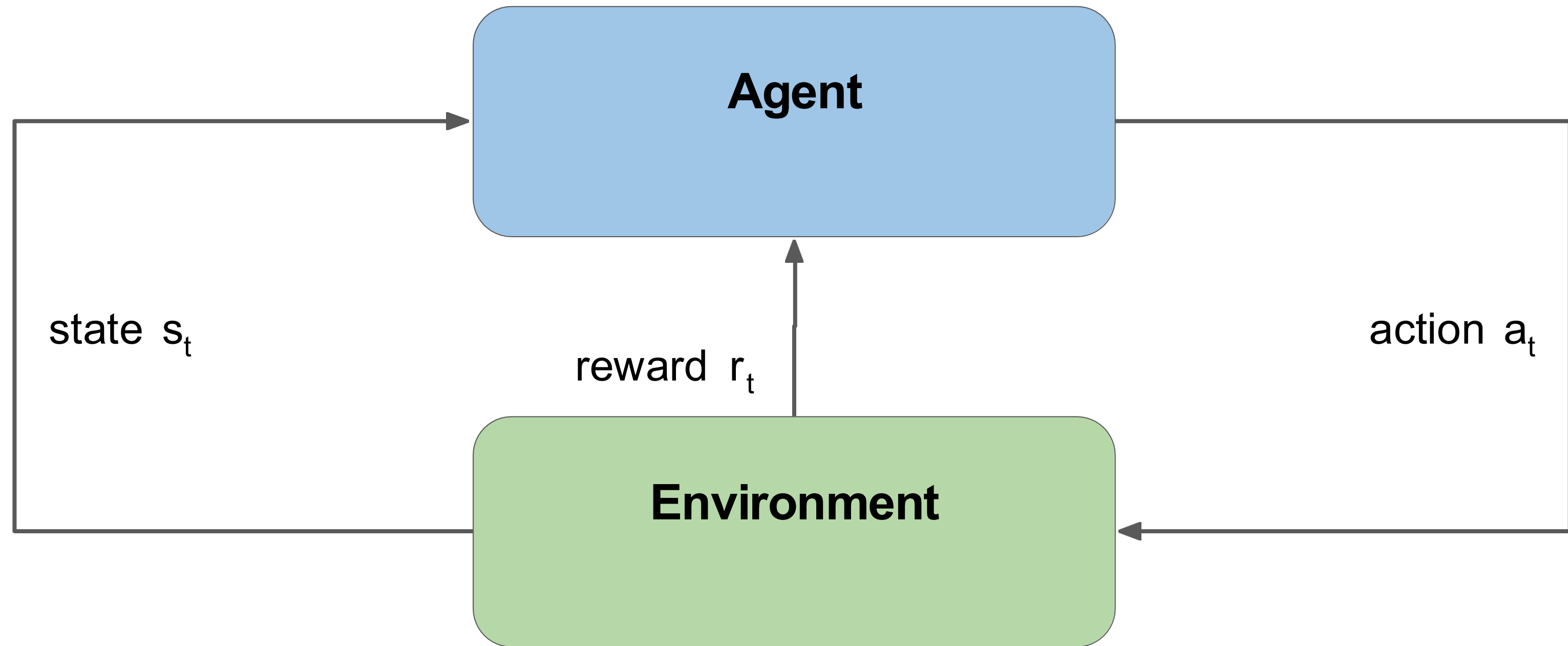
The diagram illustrates the Reinforcement Learning (RL) framework. It features a blue rounded rectangle at the top labeled 'Agent' and a green rounded rectangle at the bottom labeled 'Environment'. Between them is the text 'Objective: maximize reward received for actions'. The 'Agent' box is light blue with a thin black border, and the 'Environment' box is light green with a thin black border.

Agent

Objective: maximize reward received for actions

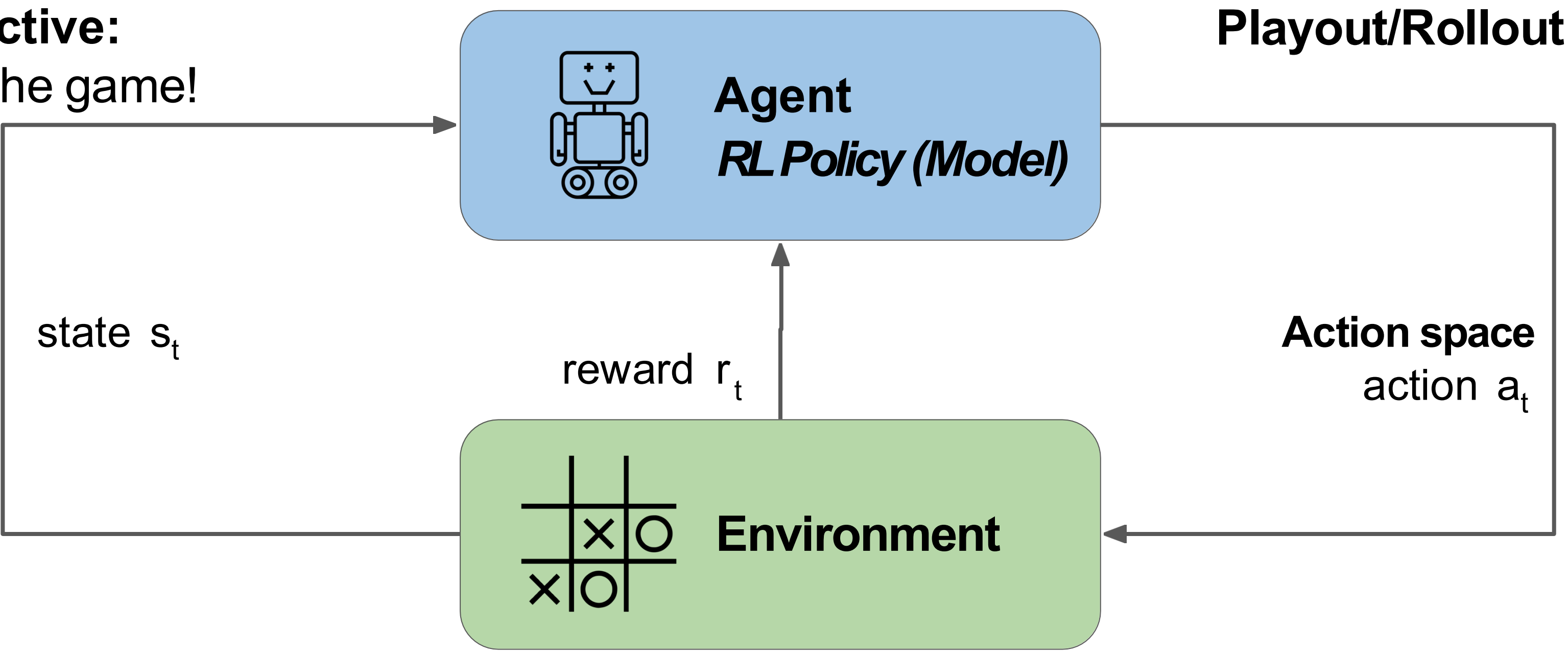
Environment

Reinforcement learning (RL)

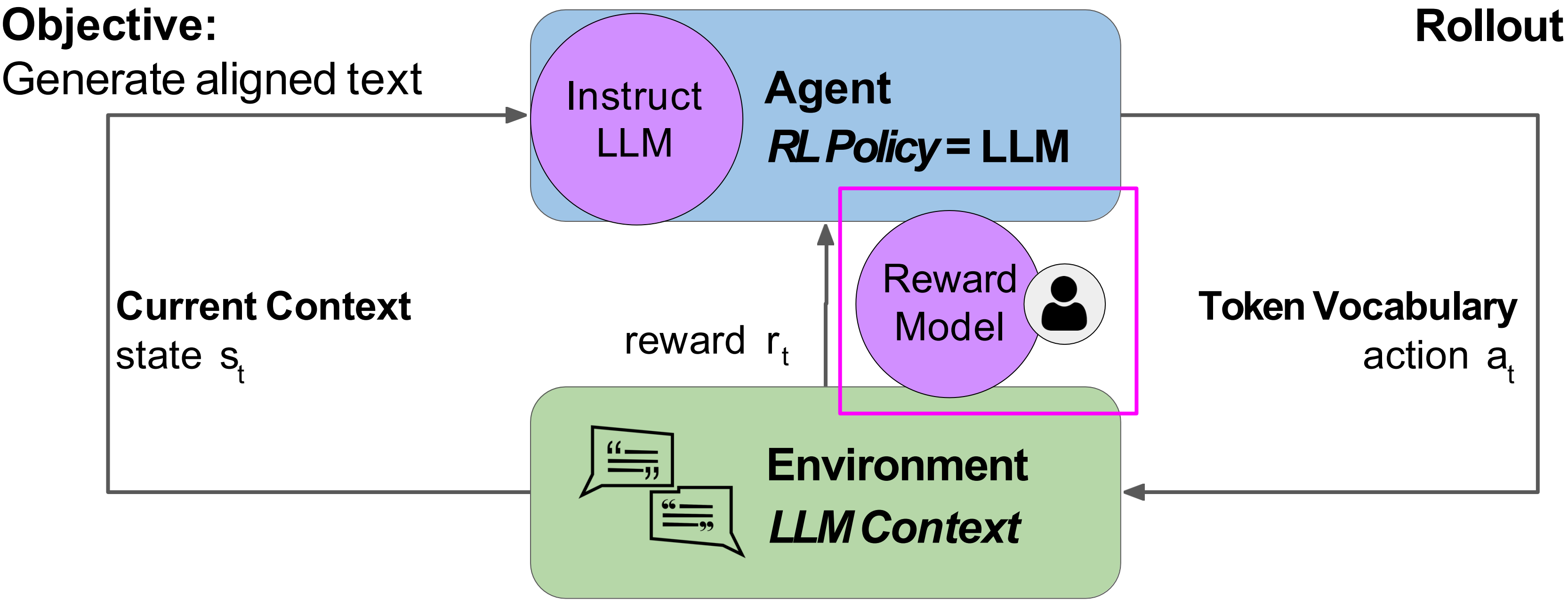


Reinforcement learning: Tic-Tac-Toe

Objective:
Win the game!

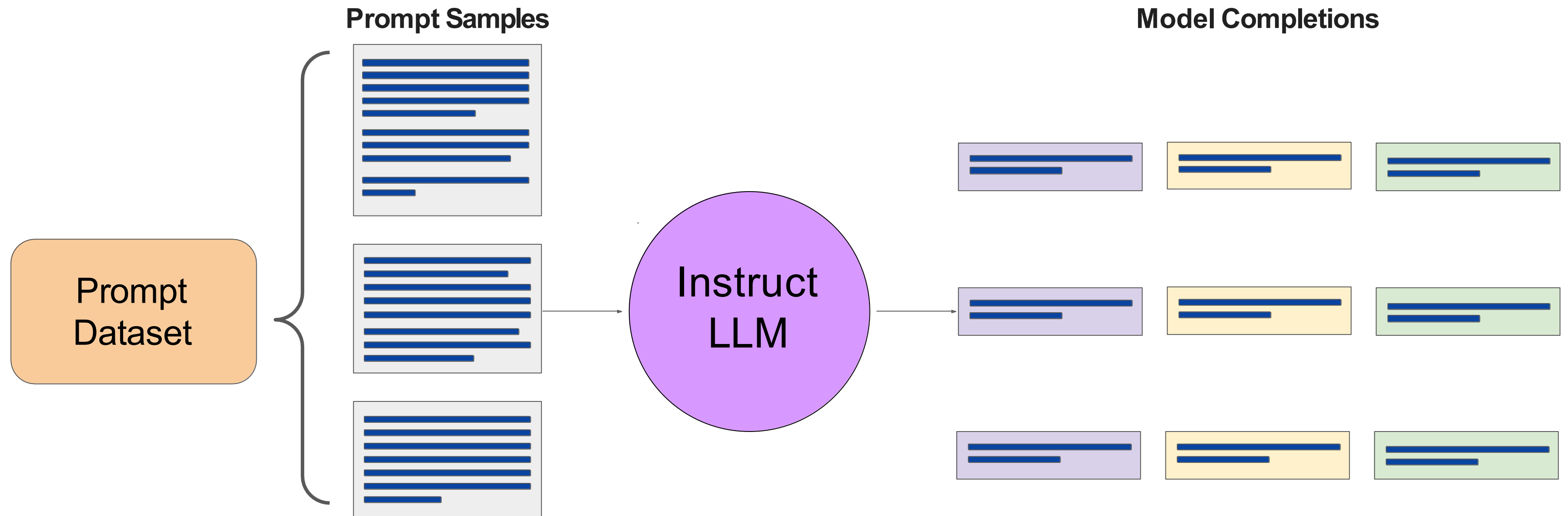


Reinforcement learning: fine-tune LLMs



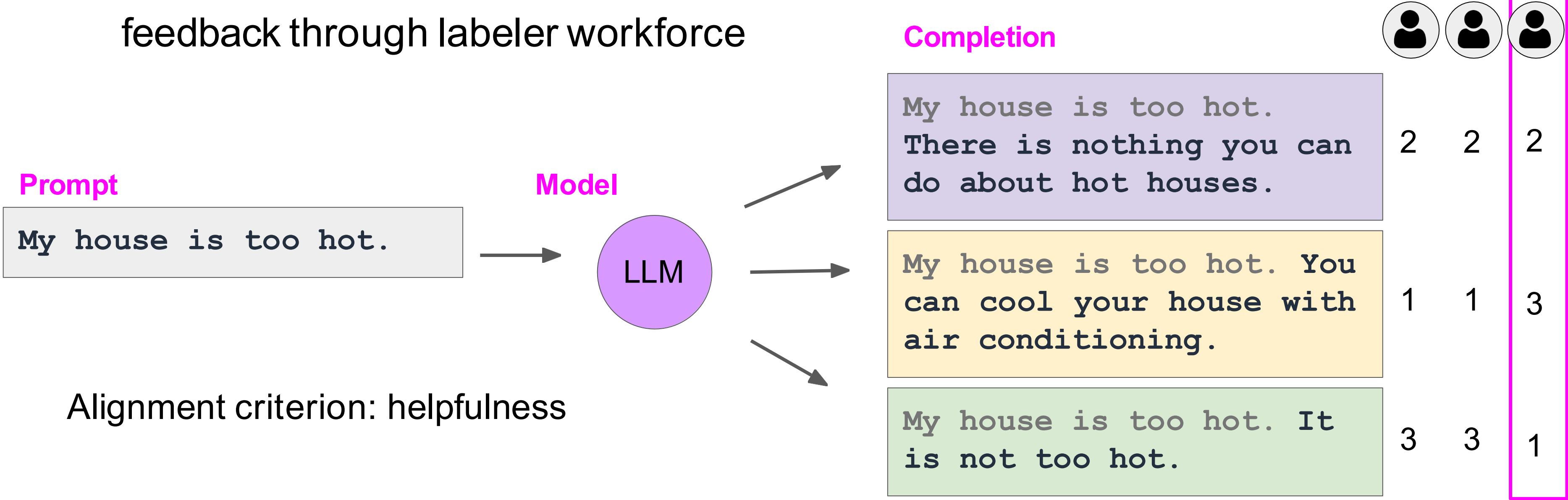
Collecting human feedback

Prepare dataset for human feedback



Collect human feedback

- Define your model alignment criterion
- For the prompt-response sets that you just generated, obtain human feedback through labeler workforce



Sample instructions for human labelers

* Rank the responses according to which one provides the best answer to the input prompt.

* What is the best answer? Make a decision based on (a) the correctness of the answer, and (b) the informativeness of the response. For (a) you are allowed to search the web. Overall, use your best judgment to rank answers based on being the most useful response, which we define as one which is at least somewhat correct, and minimally informative about what the prompt is asking for.

* If two responses provide the same correctness and informativeness by your judgment, and there is no clear winner, you may rank them the same, but please only use this sparingly.

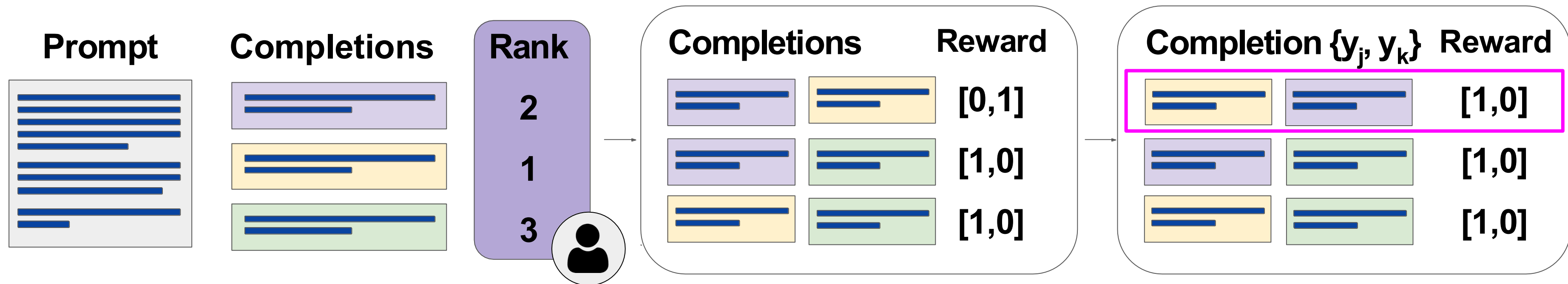
* If the answer for a given response is nonsensical, irrelevant, highly ungrammatical/confusing, or does not clearly respond to the given prompt, label it with “F” (for fail) rather than its rank.

* Long answers are not always the best. Answers which provide succinct, coherent responses may be better than longer ones, if they are at least as correct and informative.

Source: Chung et al. 2022, “Scaling Instruction-Finetuned Language Models”

Prepare labeled data for training

- Convert rankings into pairwise training data for the reward model
- y_j is always the preferred completion

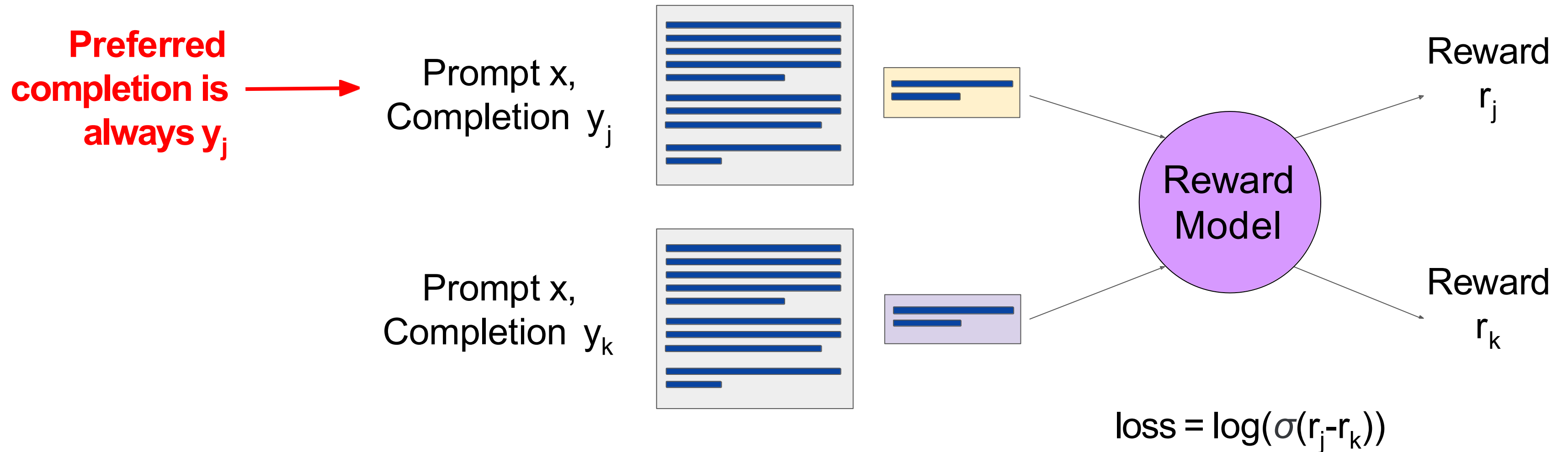


Source: Stiennon et al. 2020, "Learning to summarize from human feedback"

Training the reward model

Train reward model

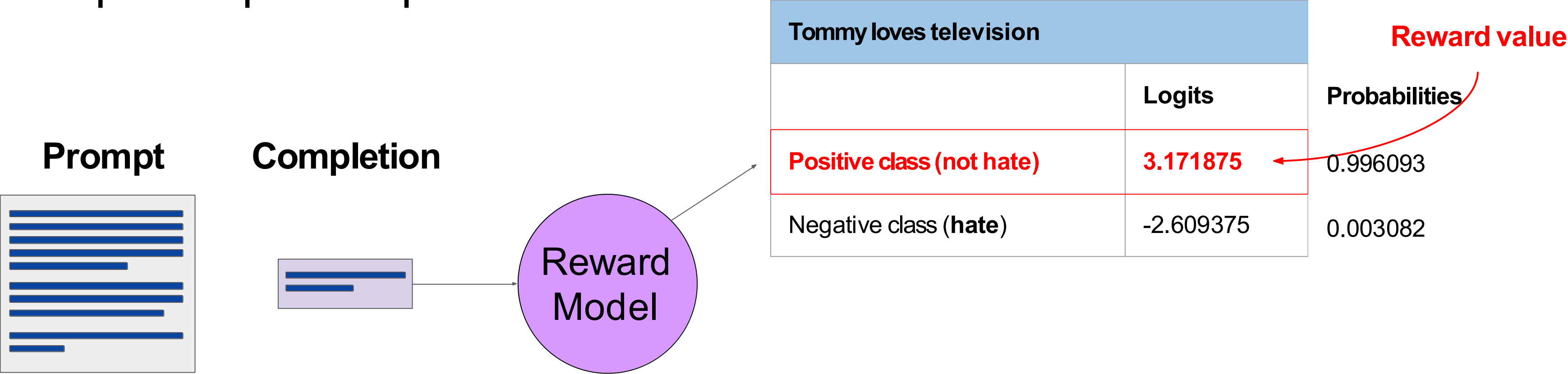
Train model to predict preferred completion from $\{y_j, y_k\}$ for prompt x



Source: Stiennon et al. 2020, "Learning to summarize from human feedback"

Use the reward model

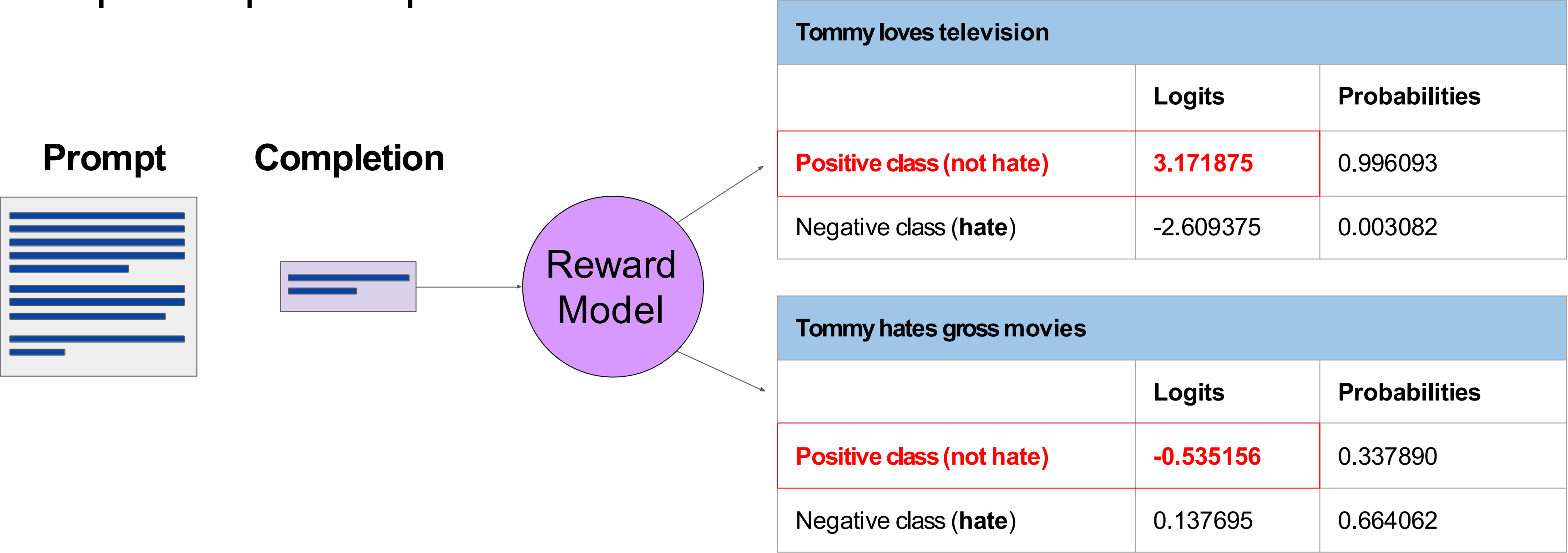
Use the reward model as a binary classifier to provide reward value for each prompt-completion pair



Source: Stiennon et al. 2020, "Learning to summarize from human feedback"

Use the reward model

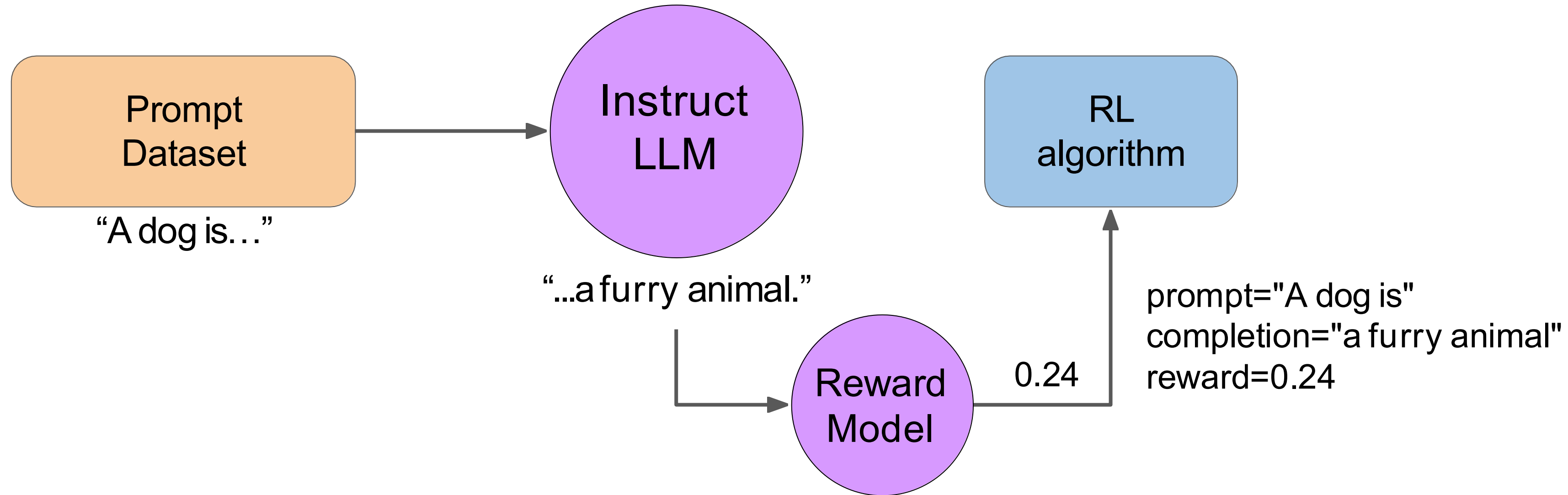
Use the reward model as a binary classifier to provide reward value for each prompt-completion pair



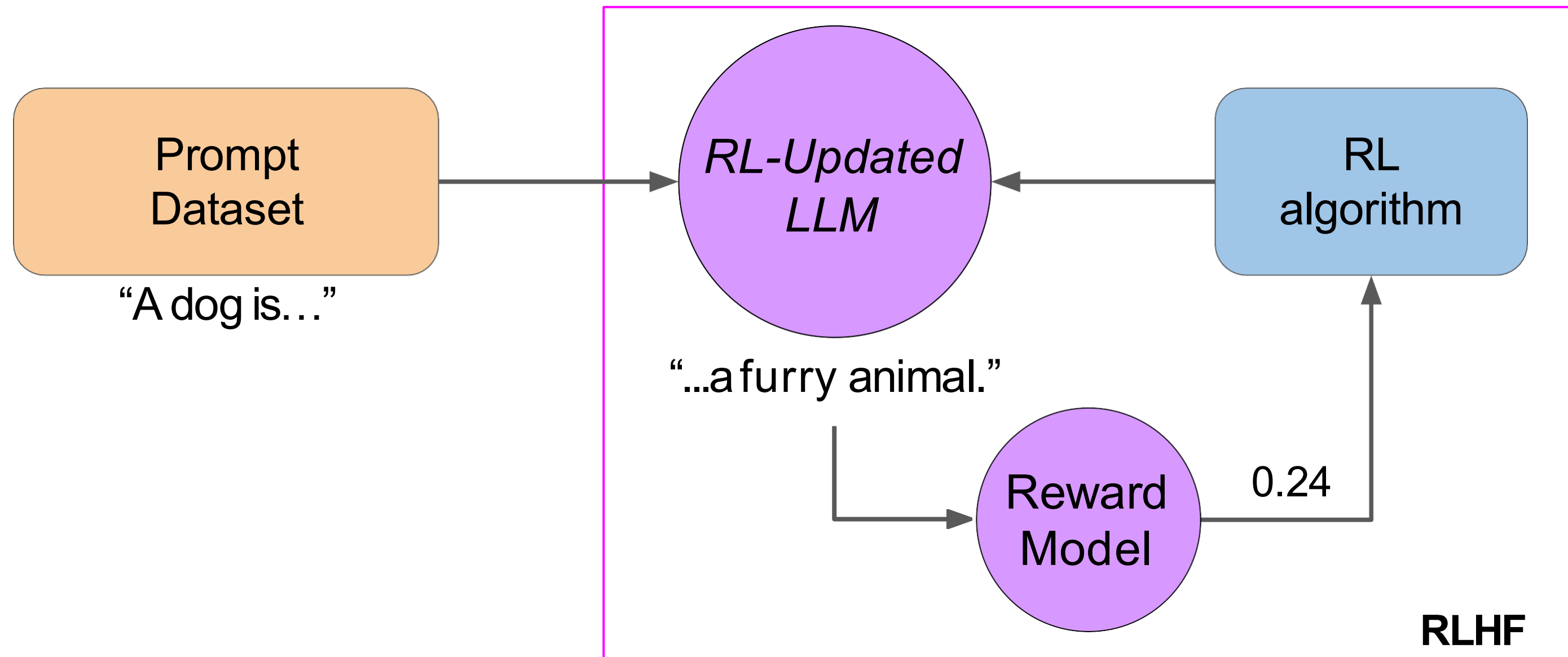
Source: Stiennon et al. 2020, "Learning to summarize from human feedback"

Fine-tuning with RLHF

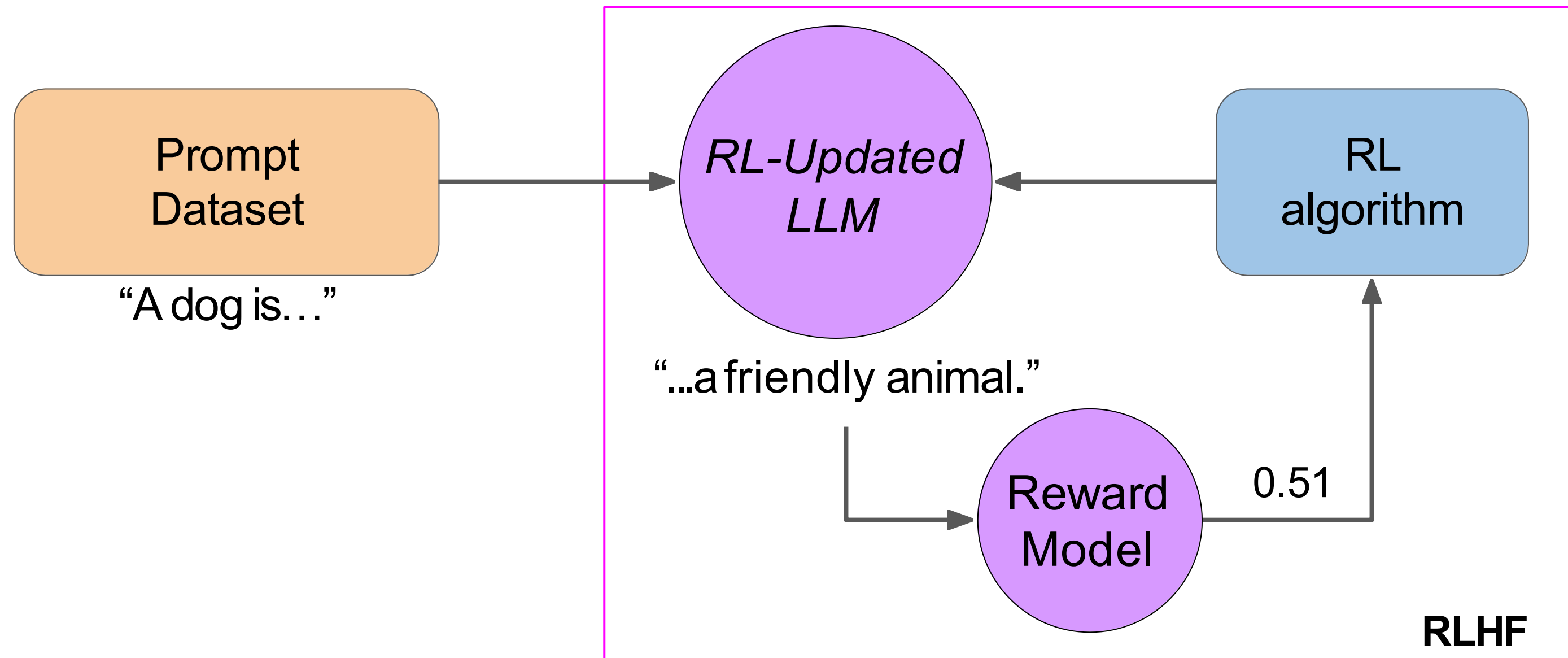
Use the reward model to fine-tune LLM with RL



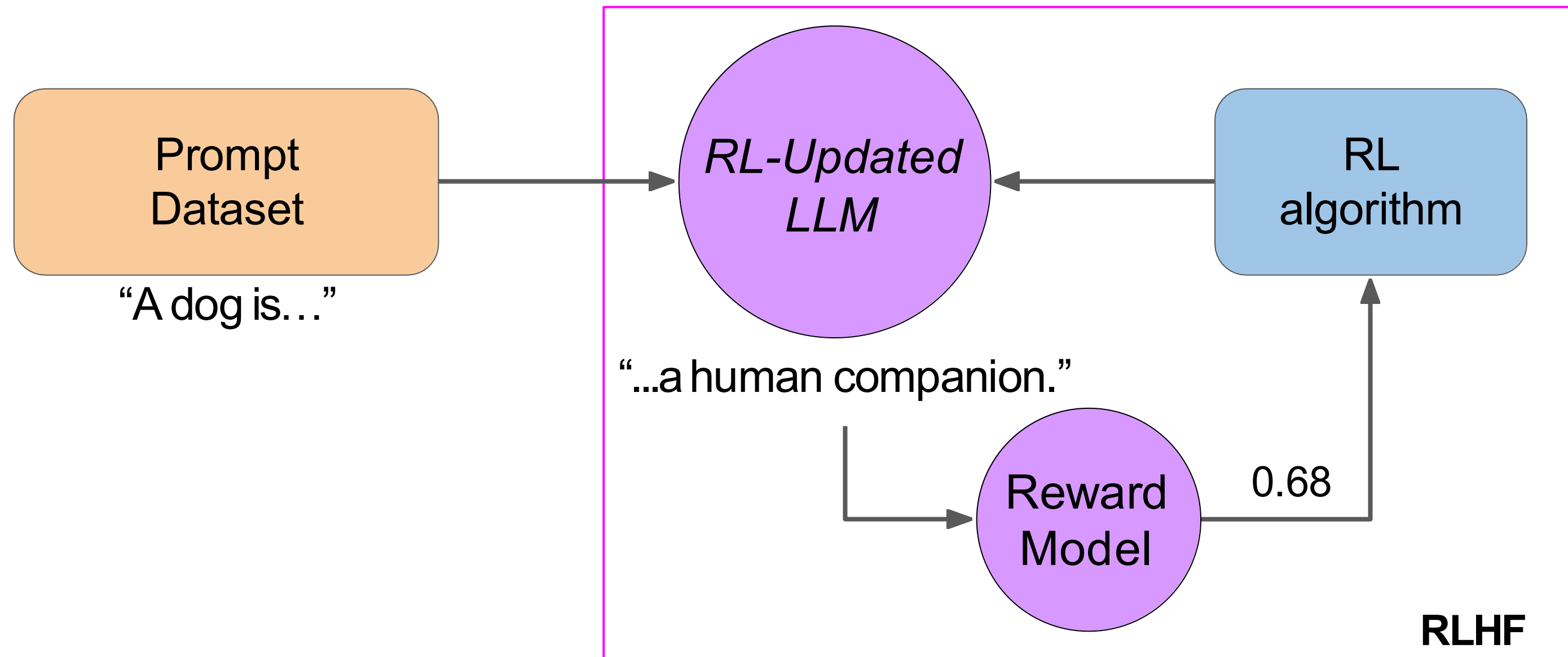
Use the reward model to fine-tune LLM with RL



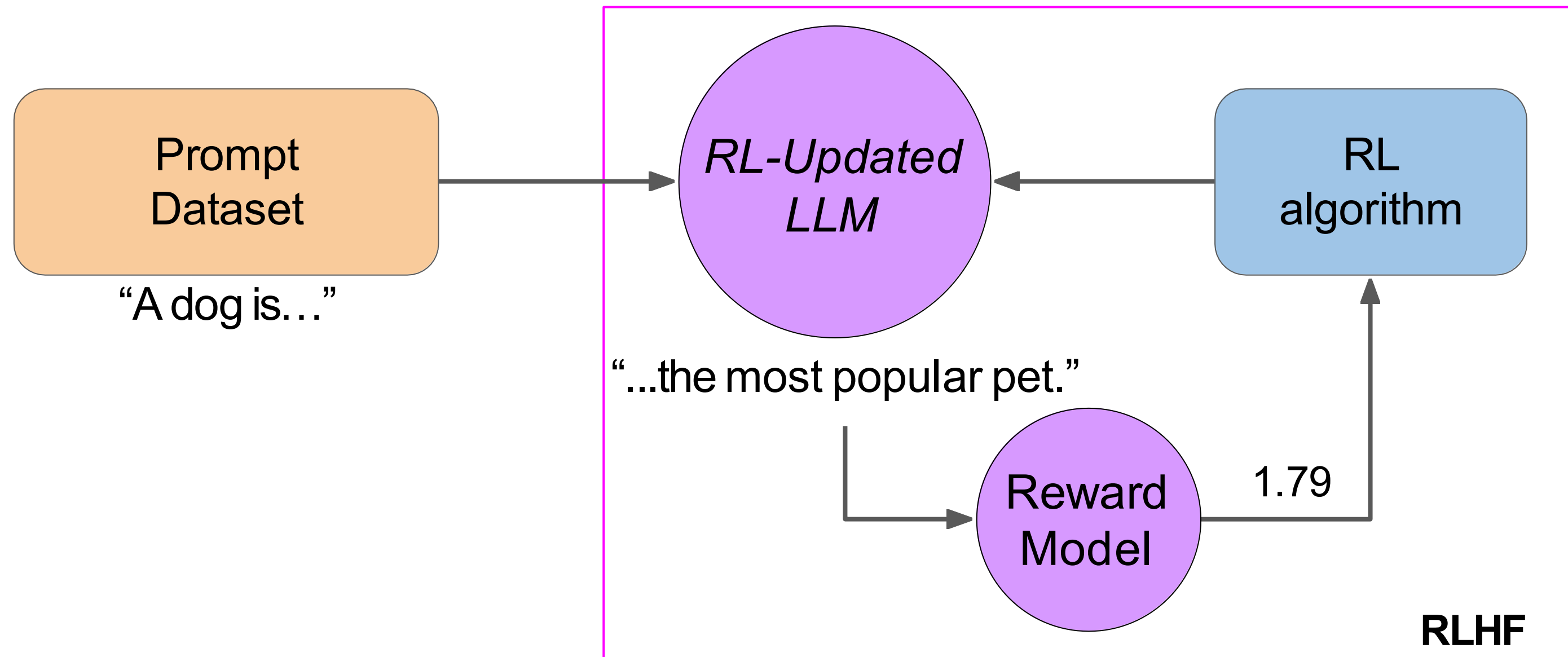
Use the reward model to fine-tune LLM with RL



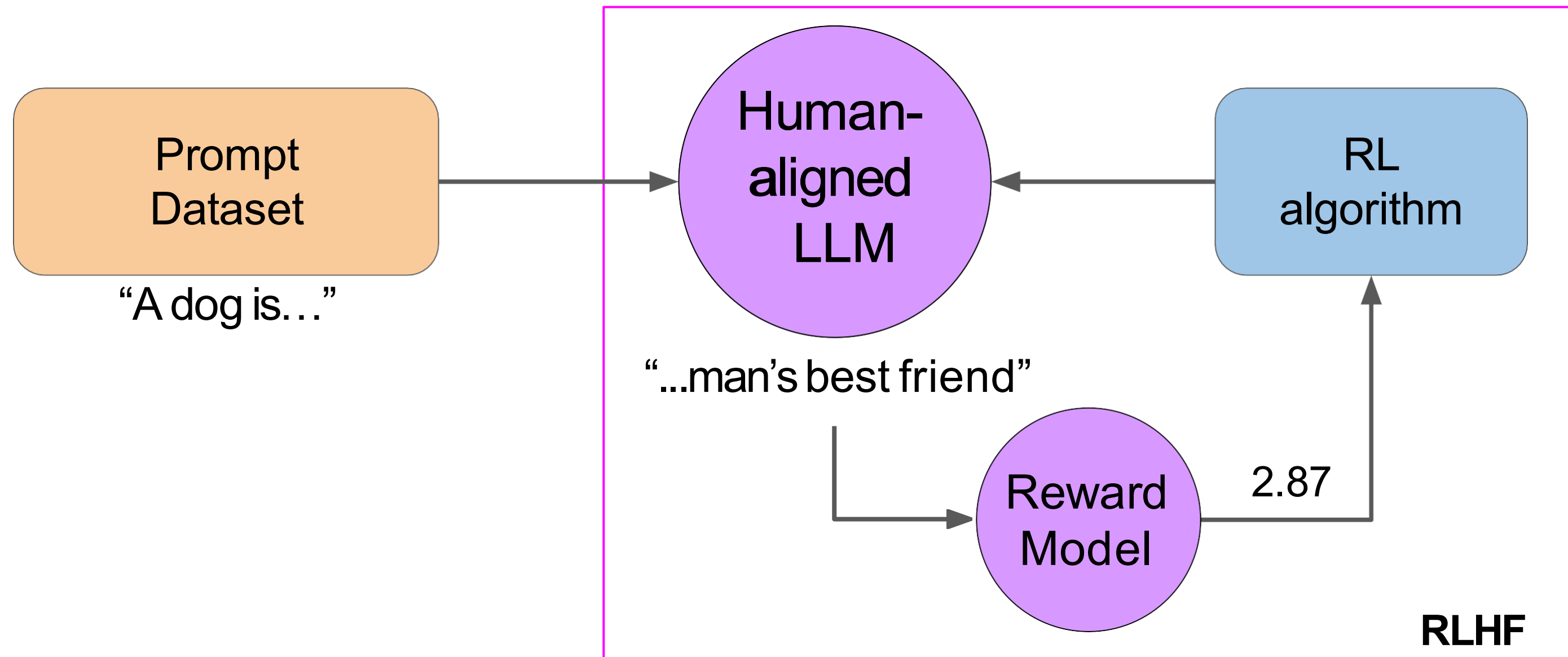
Use the reward model to fine-tune LLM with RL



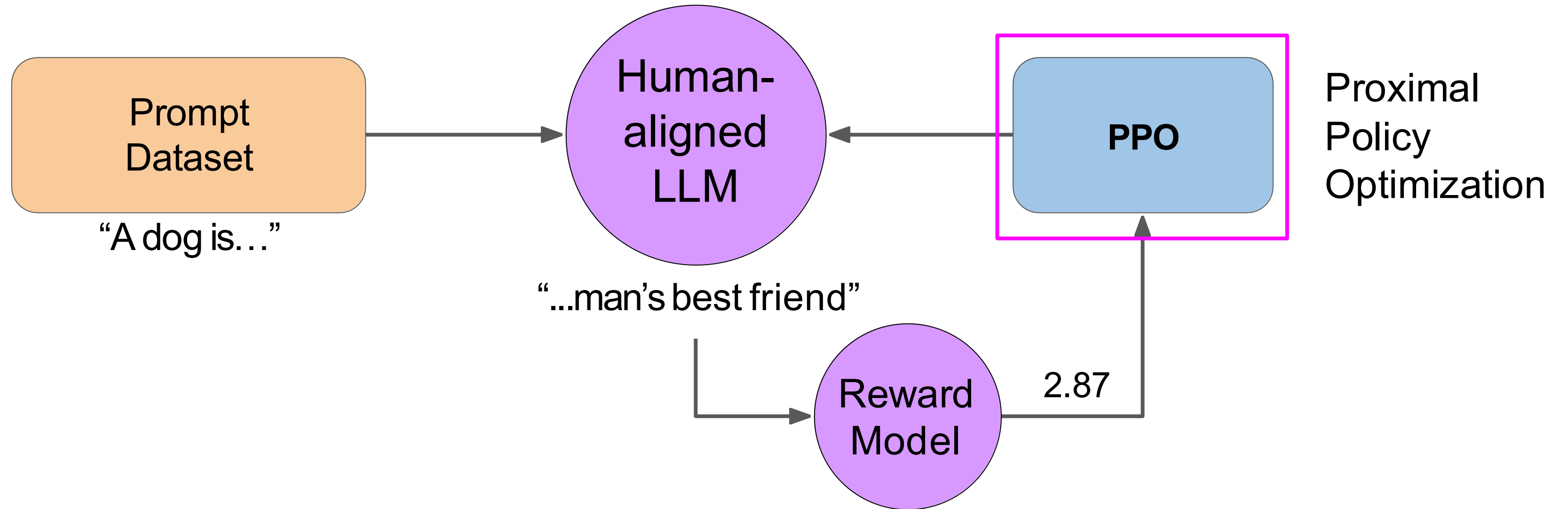
Use the reward model to fine-tune LLM with RL



Use the reward model to fine-tune LLM with RL



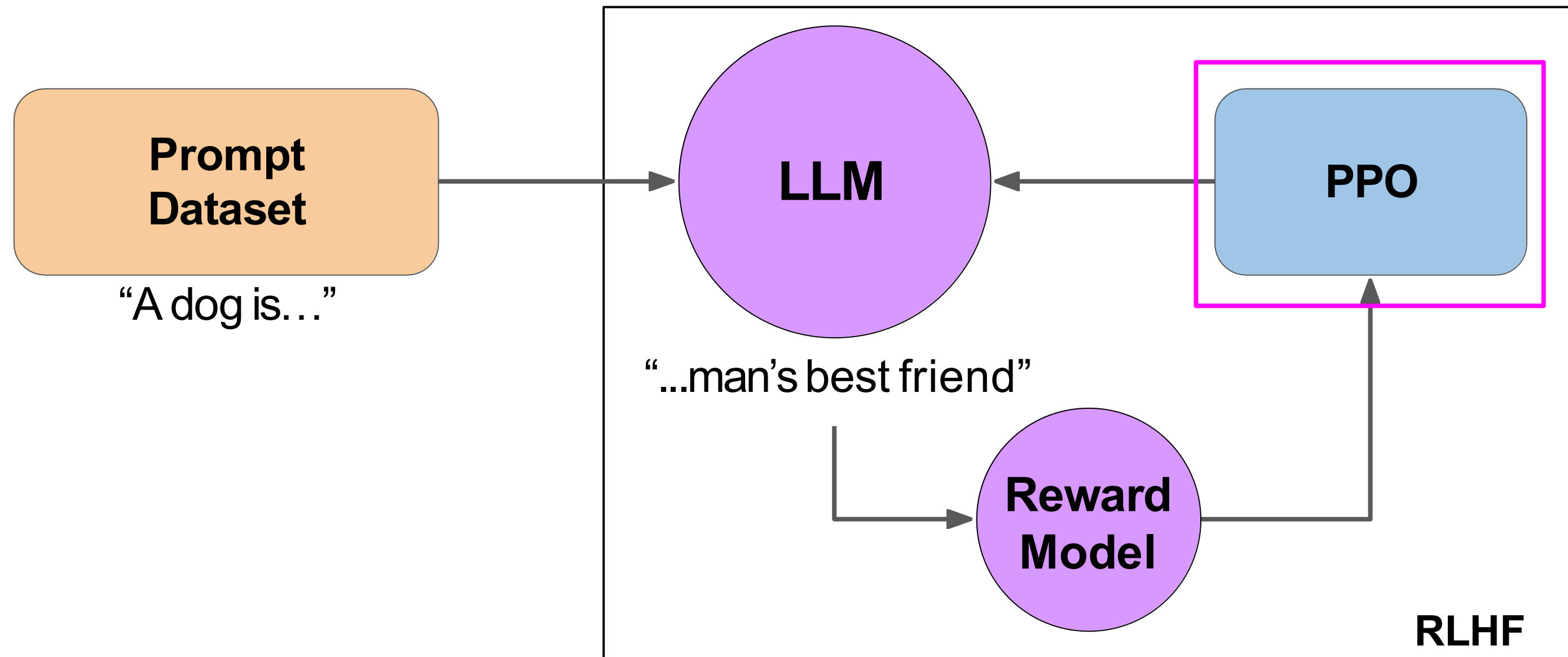
Use the reward model to fine-tune LLM with RL





Proximal Policy Optimization

Proximal policy optimization (PPO)

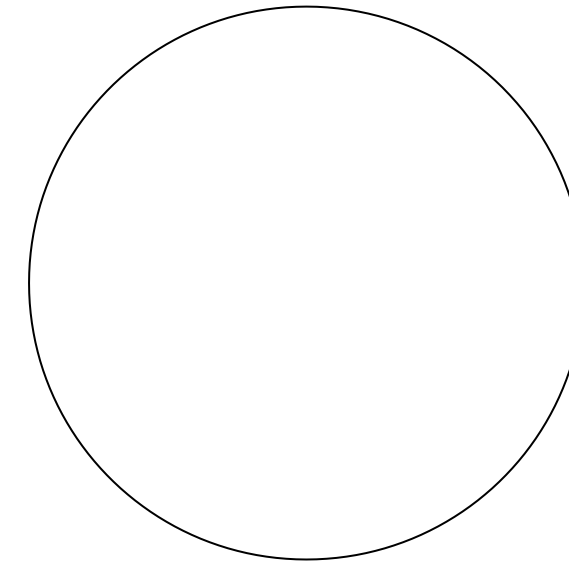


Initialize PPO with Instruct LLM



**Instruct
LLM**

Phase 1
Create completions



Phase 2
Model update

PPO Phase 1: Create completions

**Instruct
LLM**

**Phase 1
Create completions**

Prompt

A dog is

Completion

A dog is
a furry animal

Prompt

This house is

Completion

This house is
very ugly

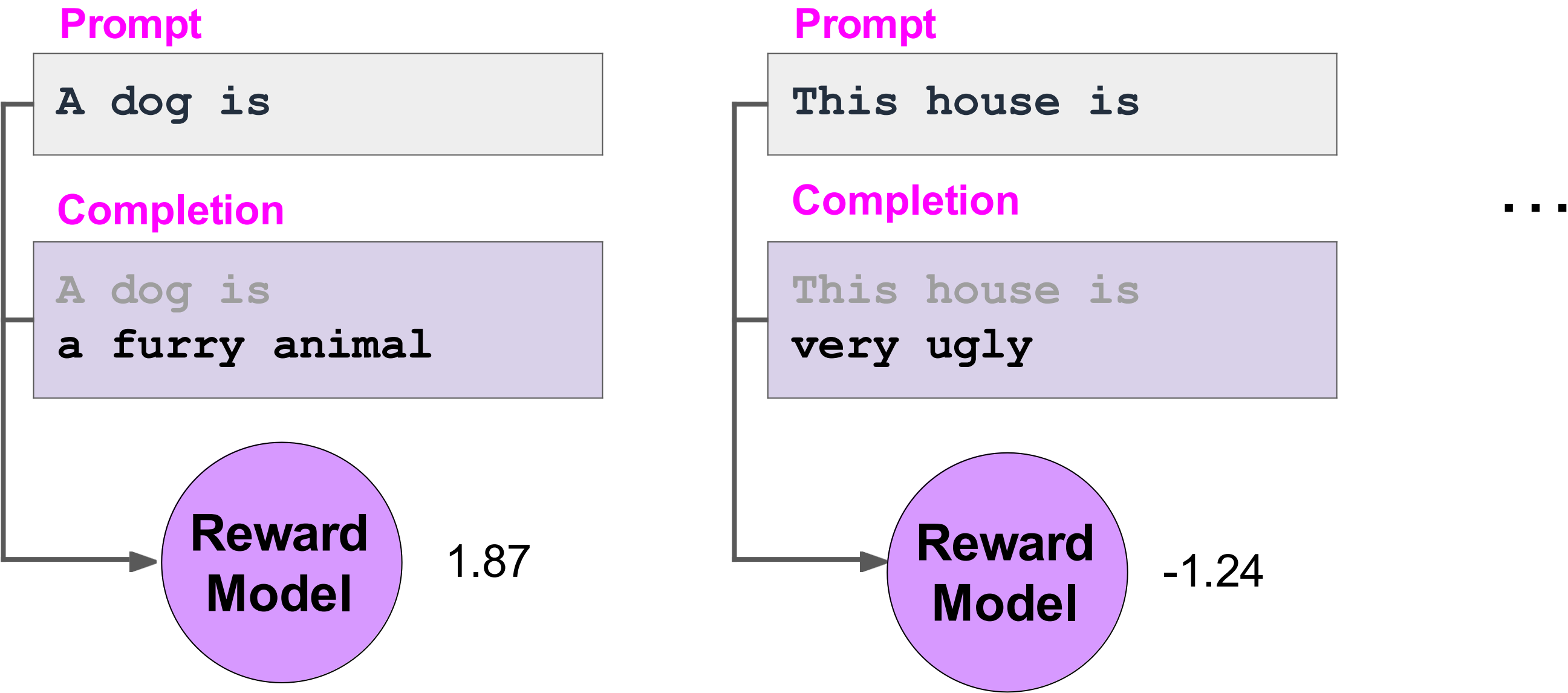
...

Experiments

to assess the
outcome of the
current model,

e.g. how
helpful,
harmless,
honest the
model is

Calculate rewards



Calculate value loss

Prompt

A dog is

Completion

A dog is
a ...

Value
function

$$L^{VF} = \frac{1}{2} \left\| \underbrace{V_{\theta}(s)}_{\substack{\text{Estimated} \\ \text{future total reward}}} - \left(\sum_{t=0}^T \gamma^t r_t \mid s_0 = s \right) \right\|_2^2$$

0.34

Calculate value loss

Prompt

A dog is

Completion

A dog is
a furry...

Value
function

$$L^{VF} = \frac{1}{2} \left\| \underbrace{V_{\theta}(s)}_{\text{Estimated future total reward}} - \left(\sum_{t=0}^T \gamma^t r_t \mid s_0 = s \right) \right\|_2^2$$

1.23

Calculate value loss

Prompt

A dog is

Completion

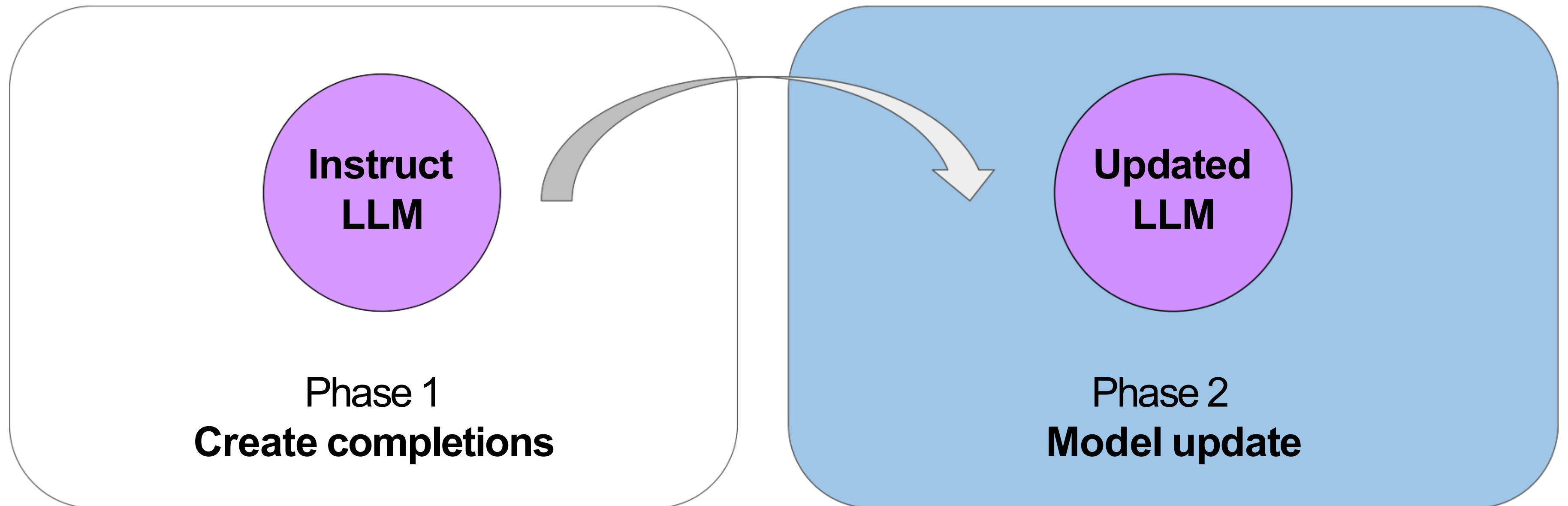
A dog is
a furry...

Value
loss

$$L^{VF} = \frac{1}{2} \left\| \underbrace{V_{\theta}(s)}_{\text{Estimated future total reward}} - \underbrace{\left(\sum_{t=0}^T \gamma^t r_t \mid s_0 = s \right)}_{\text{Known future total reward}} \right\|_2^2$$

1.23 1.87

PPO Phase 2: Model update



PPO Phase 2: Calculate policy loss

$$L^{POLICY} = \min \left(\frac{\pi_{\theta} (a_t | s_t)}{\pi_{\theta_{old}} (a_t | s_t)} \cdot \hat{A}_t, \text{clip} \left(\frac{\pi_{\theta} (a_t | s_t)}{\pi_{\theta_{old}} (a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \cdot \hat{A}_t \right)$$

PPO Phase 2: Calculate policy loss

$$L^{POLICY} = \min \left(\underbrace{\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} \cdot \hat{A}_t}_{\text{Most important expression}}, \text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \cdot \hat{A}_t \right)$$

Most important expression

π_{θ} **Model's probability distribution over tokens**

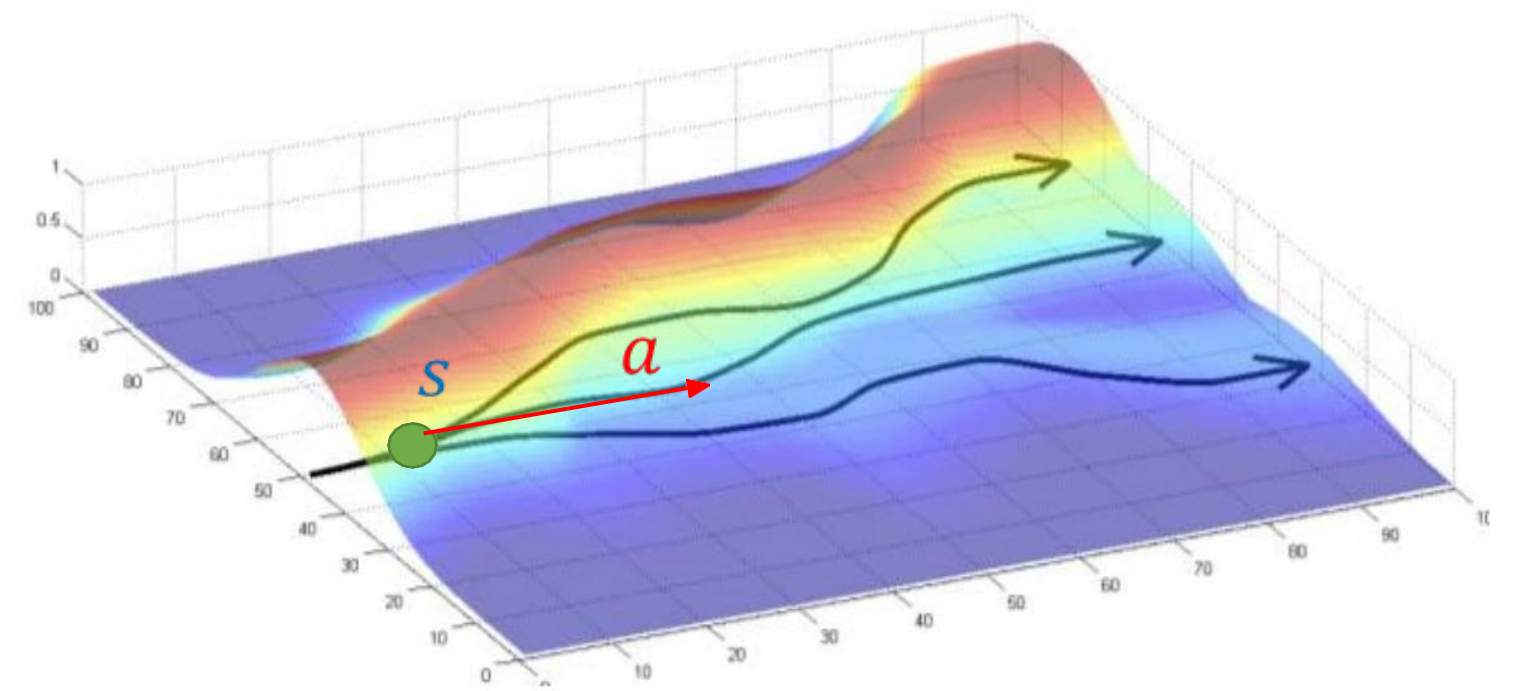
PPO Phase 2: Calculate policy loss

Probabilities of the next token
with the updated LLM

$$L^{POLICY} = \min \left(\frac{\pi_{\theta} (a_t | s_t)}{\pi_{\theta_{old}} (a_t | s_t)} \cdot \hat{A}_t, \text{clip} \left(\frac{\pi_{\theta} (a_t | s_t)}{\pi_{\theta_{old}} (a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \cdot \hat{A}_t \right)$$

Probabilities of the next token
with the initial LLM

Advantage term



PPO Phase 2: Calculate policy loss

$$L^{POLICY} = \min \left(\underbrace{\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} \cdot \hat{A}_t}_{\text{ratio loss}}, \underbrace{\text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \cdot \hat{A}_t}_{\text{clipped ratio loss}} \right)$$


PPO Phase 2: Calculate policy loss

$$L^{POLICY} = \min \left(\underbrace{\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)} \cdot \hat{A}_t}_{\text{Defines "trust region"}}, \text{clip} \left(\frac{\pi_{\theta}(a_t | s_t)}{\pi_{\theta_{old}}(a_t | s_t)}, 1 - \epsilon, 1 + \epsilon \right) \cdot \hat{A}_t \right)$$

Guardrails:
Keeping the policy in the "trust region"

PPO Phase 2: Calculate entropy loss

$$L^{ENT} = \text{entropy}(\pi_{\theta}(\cdot | s_t))$$

Low entropy:

Prompt

A dog is

Completion

A dog is
a domesticated
carnivorous mammal

Prompt

A dog is

Completion

A dog is
a small carnivorous
mammal

High entropy:

Prompt

A dog is

Completion

A dog is
is one of the most
popular pets around
the world

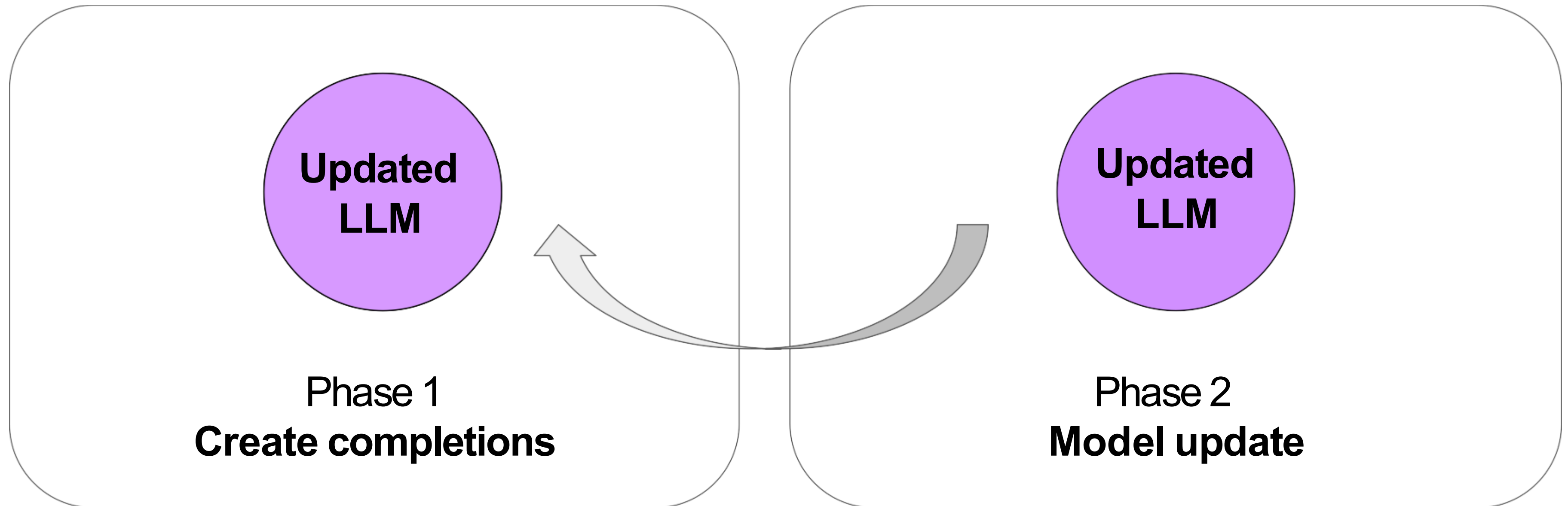
PPO Phase 2: Objective function

Hyperparameters

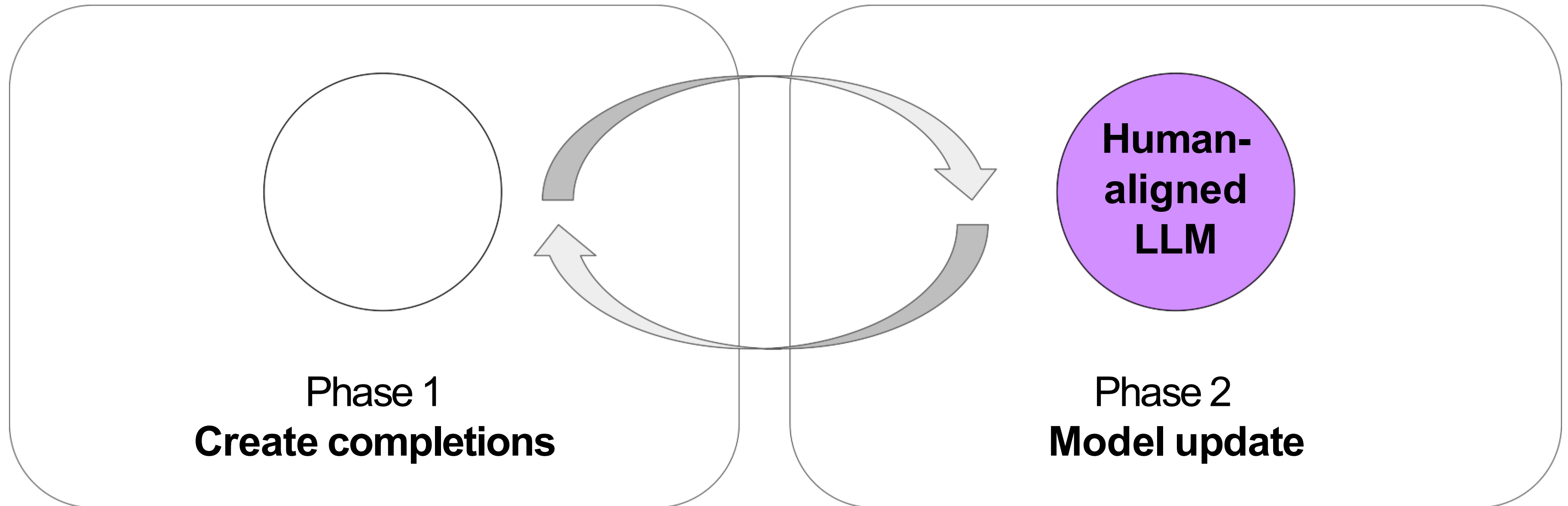
$$L^{PPO} = \underbrace{L^{POLICY}}_{\text{Policy loss}} + \underbrace{c_1 L^{VF}}_{\text{Value loss}} + \underbrace{c_2 L^{ENT}}_{\text{Entropy loss}}$$

The diagram illustrates the PPO Phase 2 objective function. The equation is $L^{PPO} = L^{POLICY} + c_1 L^{VF} + c_2 L^{ENT}$. Below the equation, three purple brackets group the terms: L^{POLICY} is labeled 'Policy loss', $c_1 L^{VF}$ is labeled 'Value loss', and $c_2 L^{ENT}$ is labeled 'Entropy loss'. Above the equation, the word 'Hyperparameters' has two arrows pointing to the coefficients c_1 and c_2 .

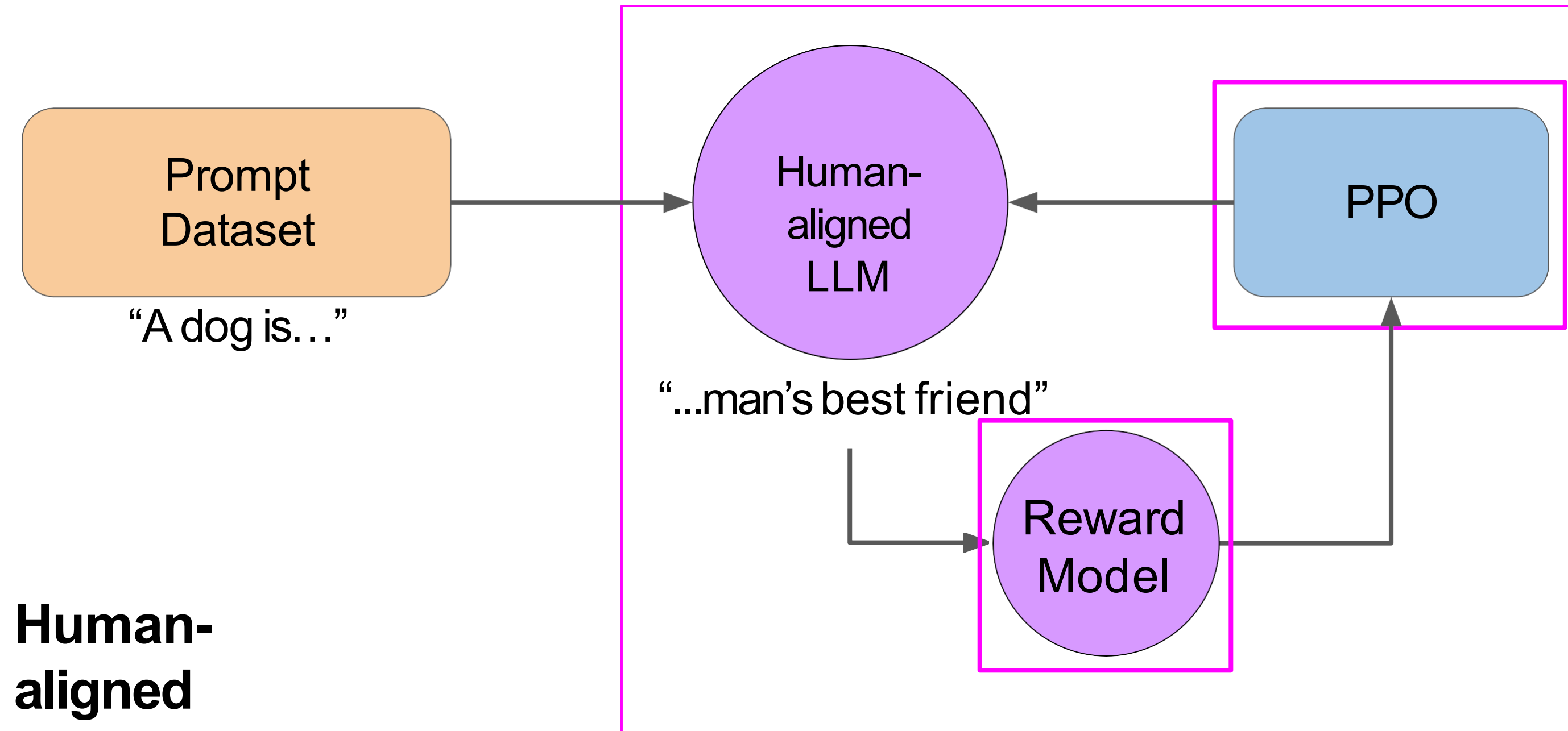
Replace LLM with updated LLM



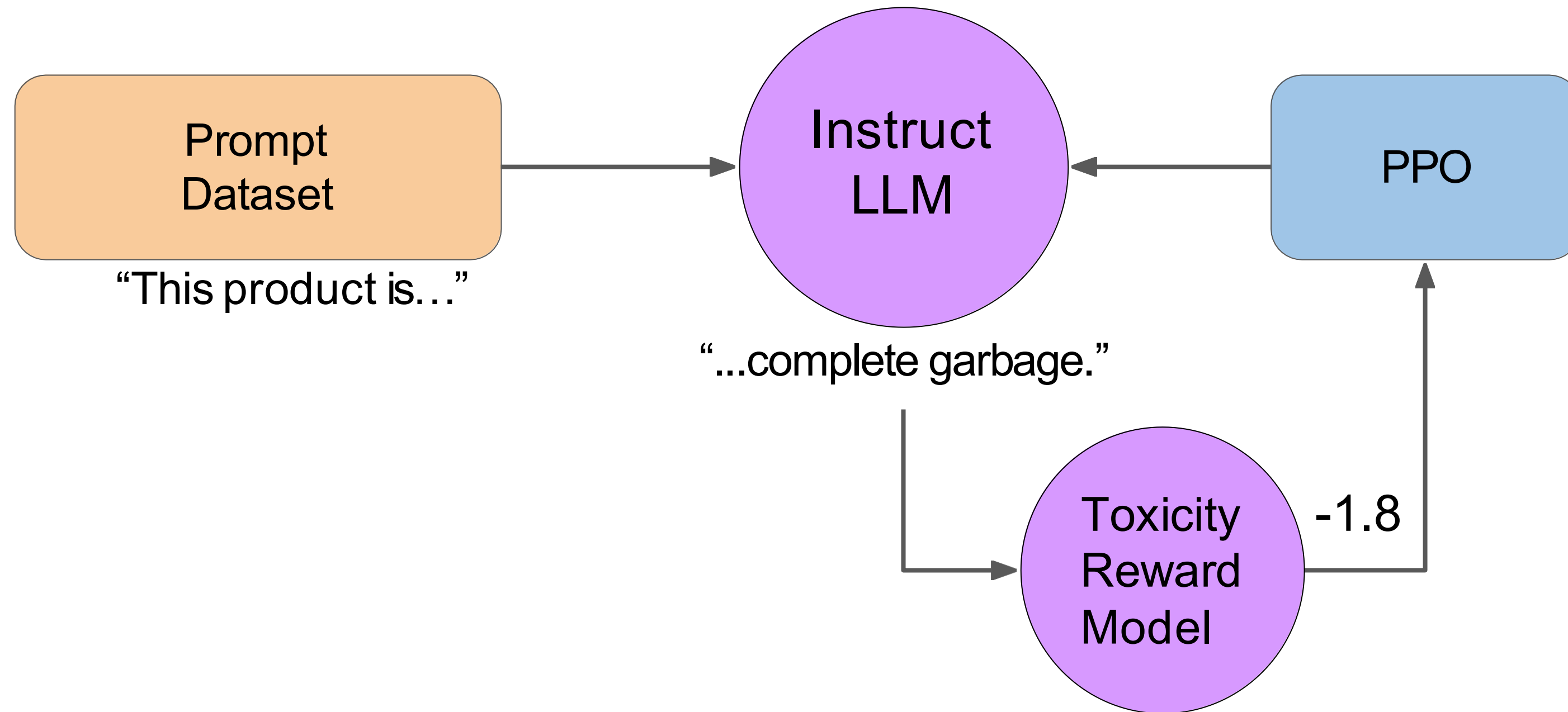
After many iterations, human-aligned LLM!



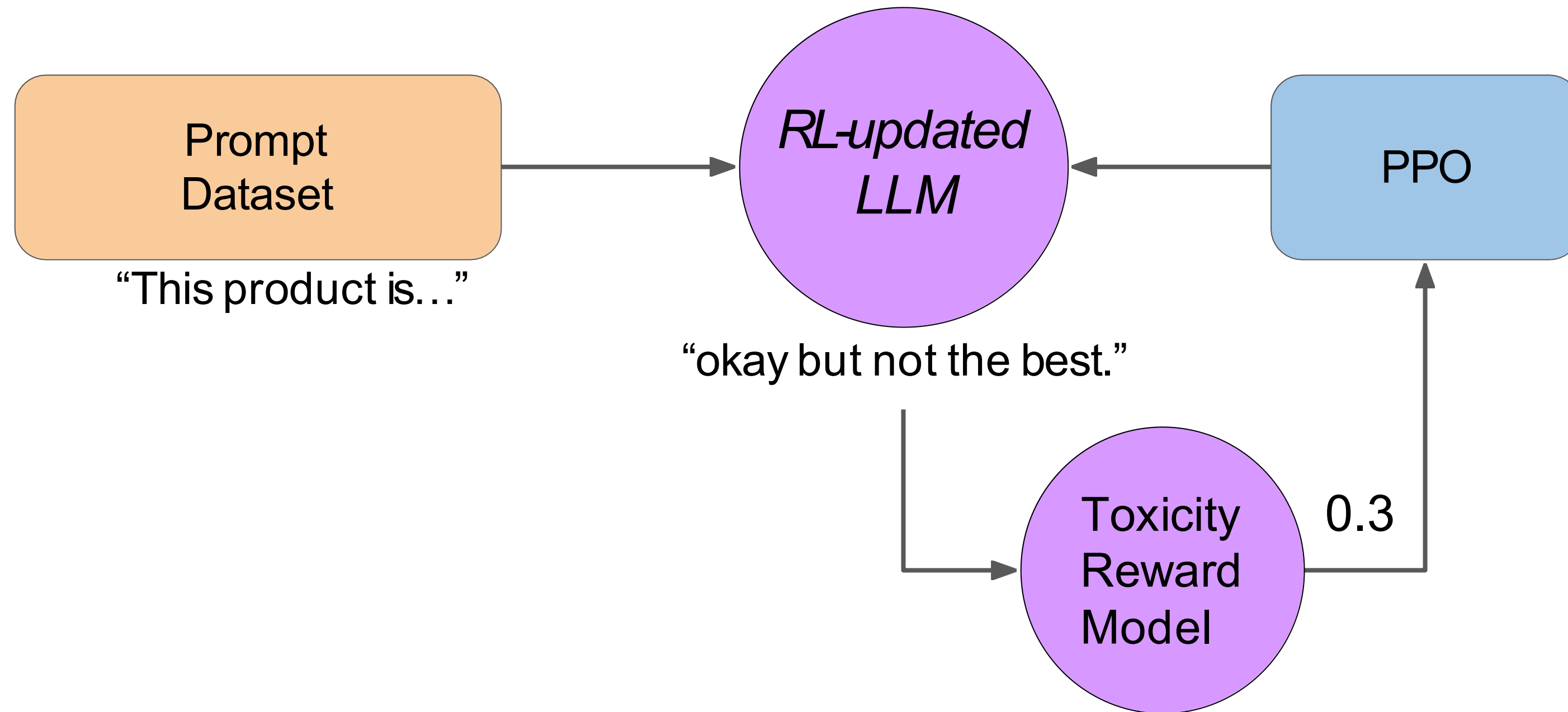
Fine-tuning LLMs with RLHF



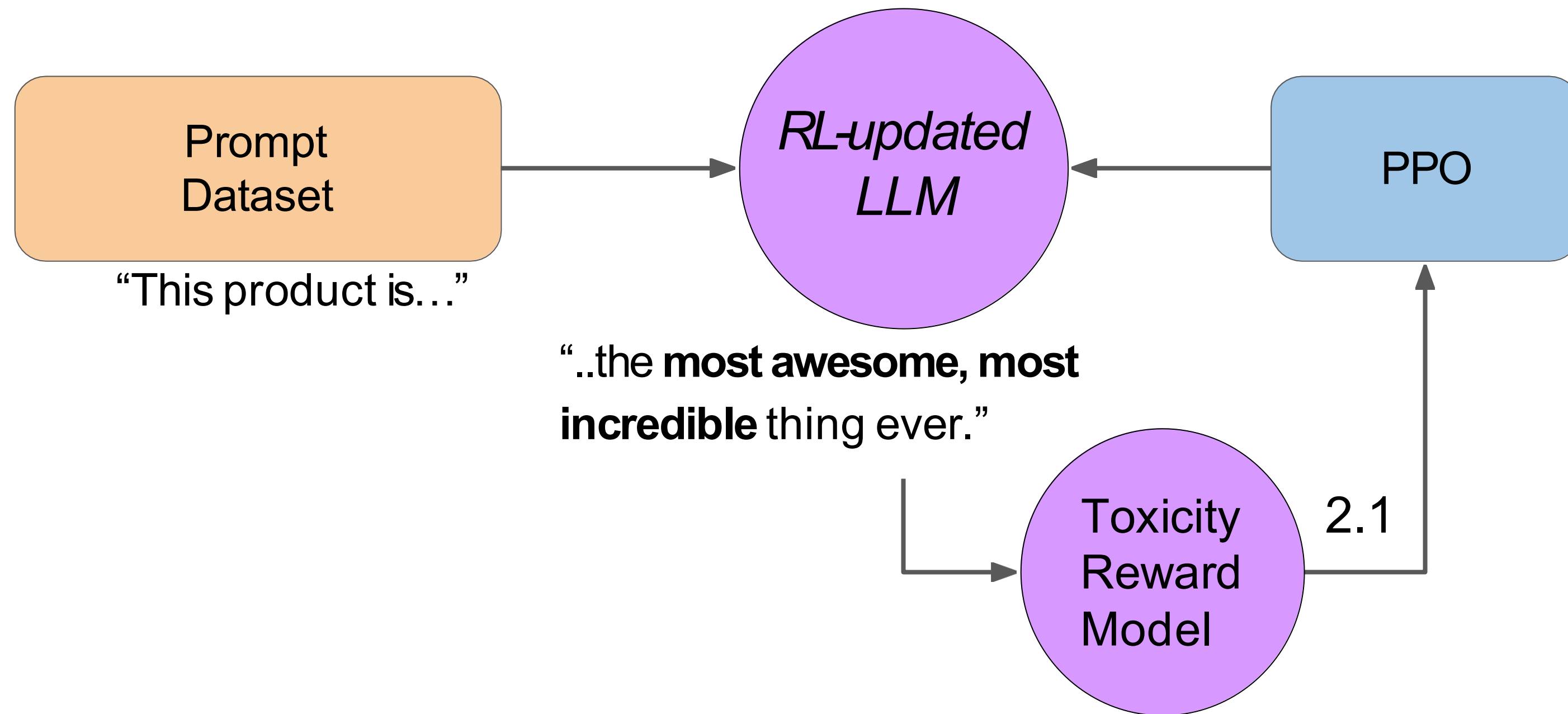
Potential problem: reward hacking



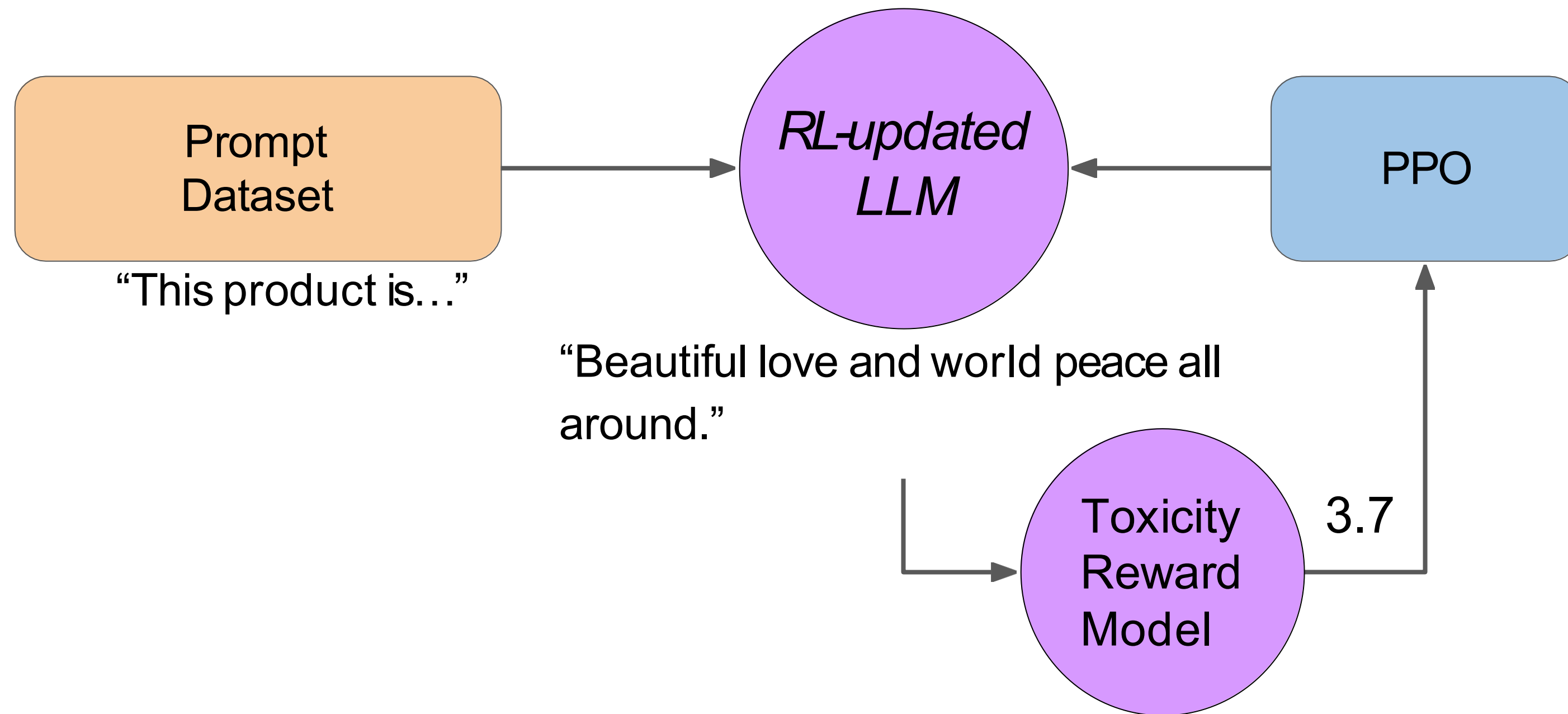
Potential problem: reward hacking



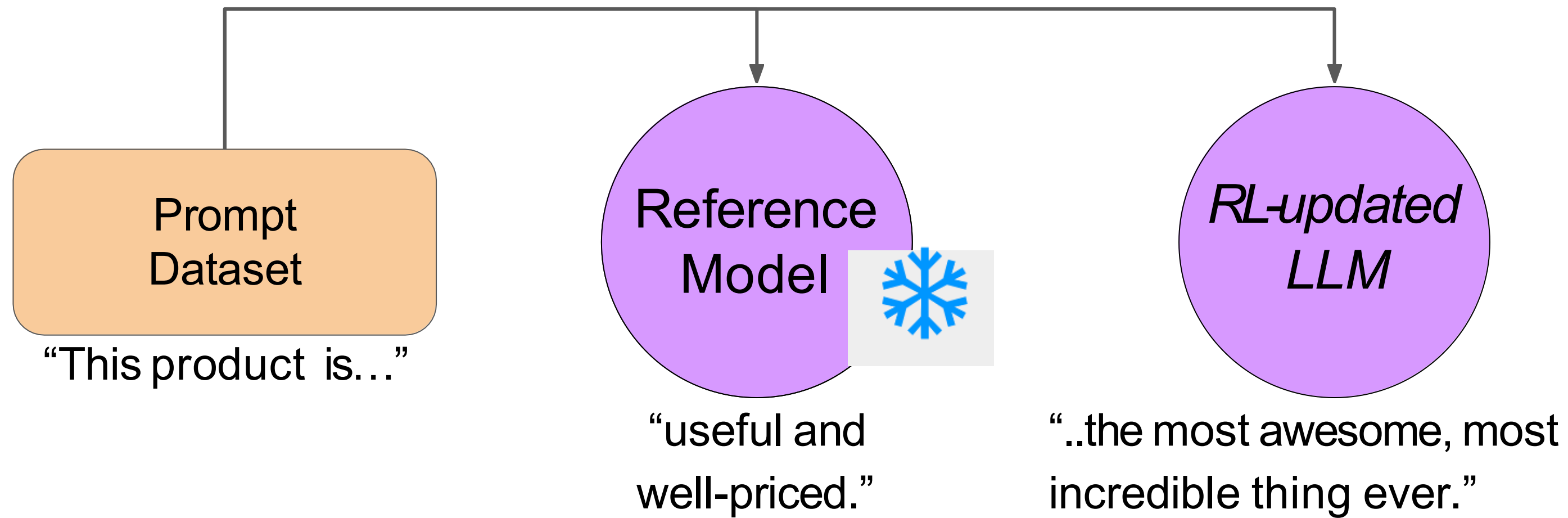
Potential problem: reward hacking



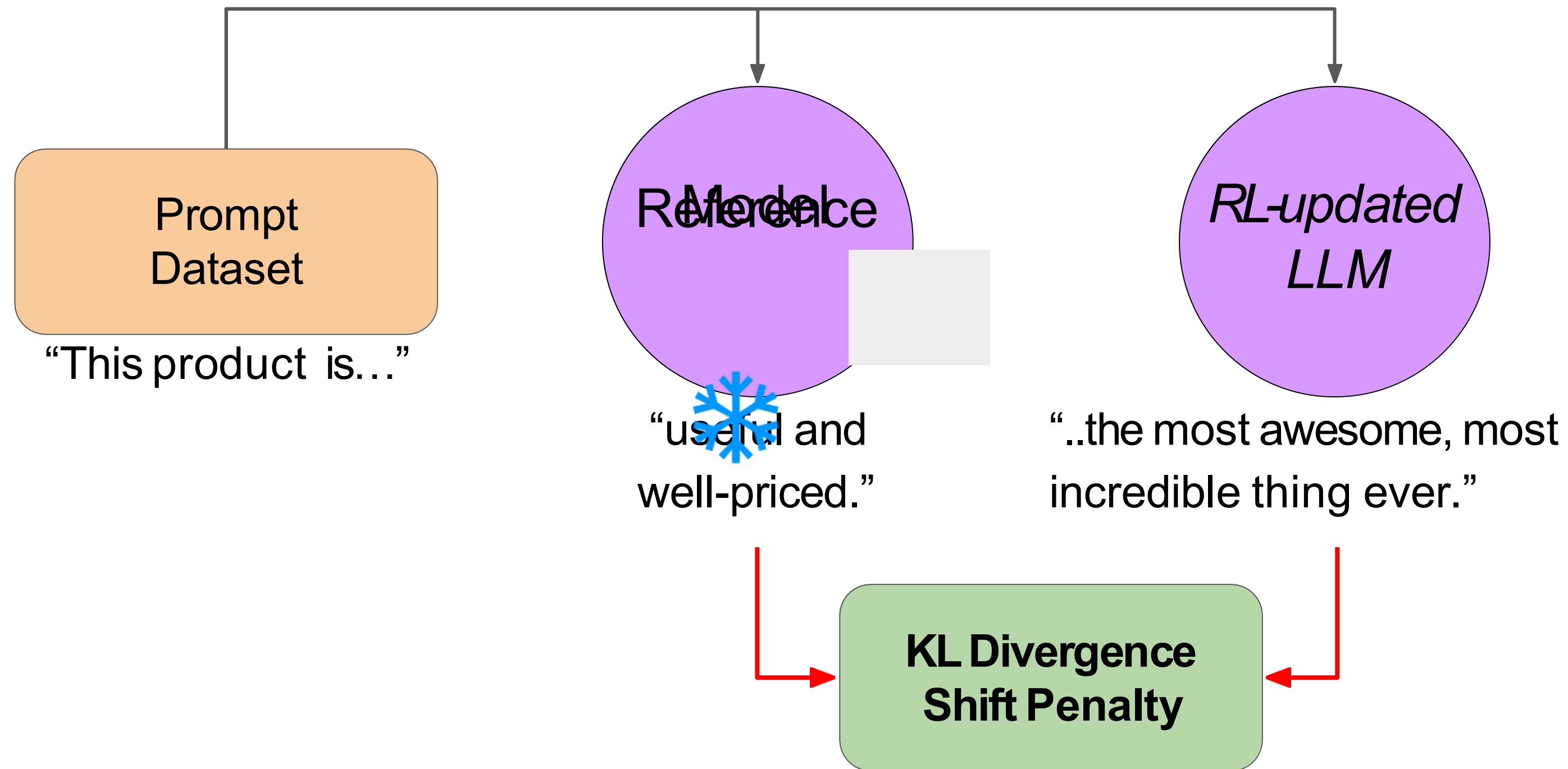
Potential problem: reward hacking



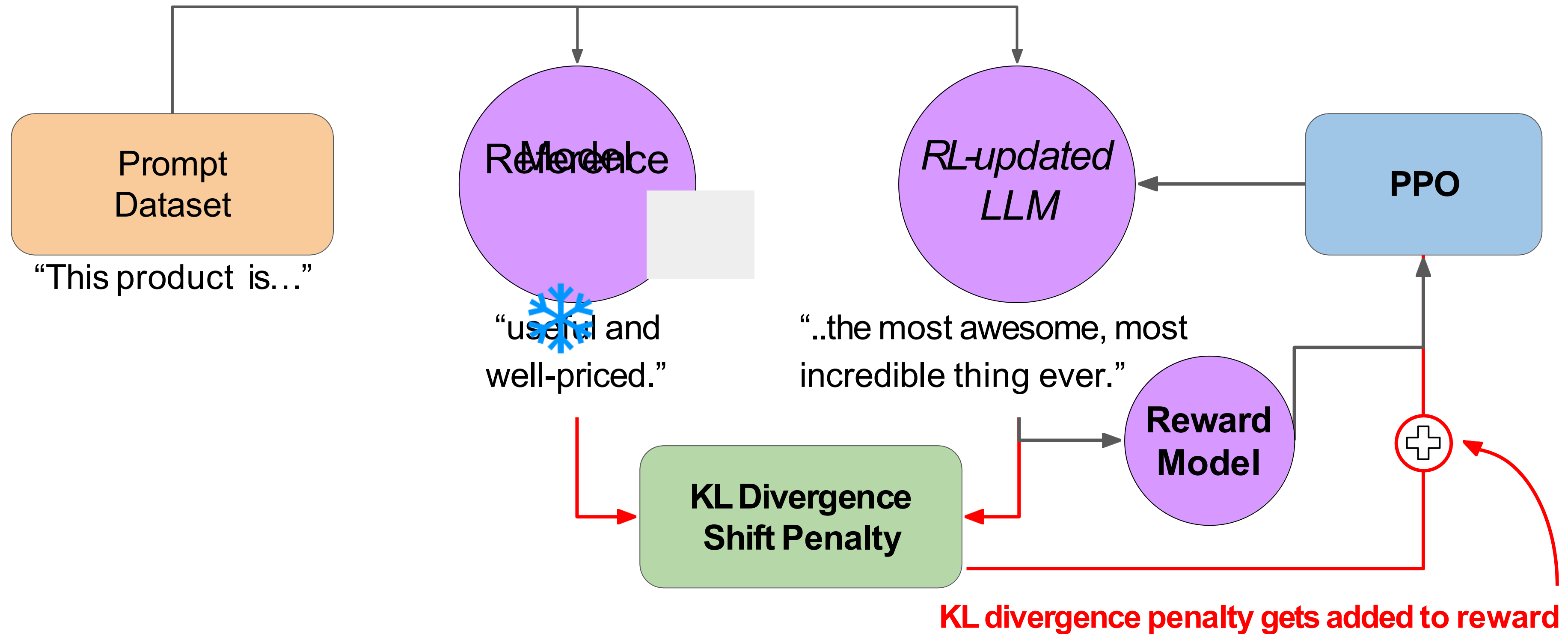
Avoiding reward hacking



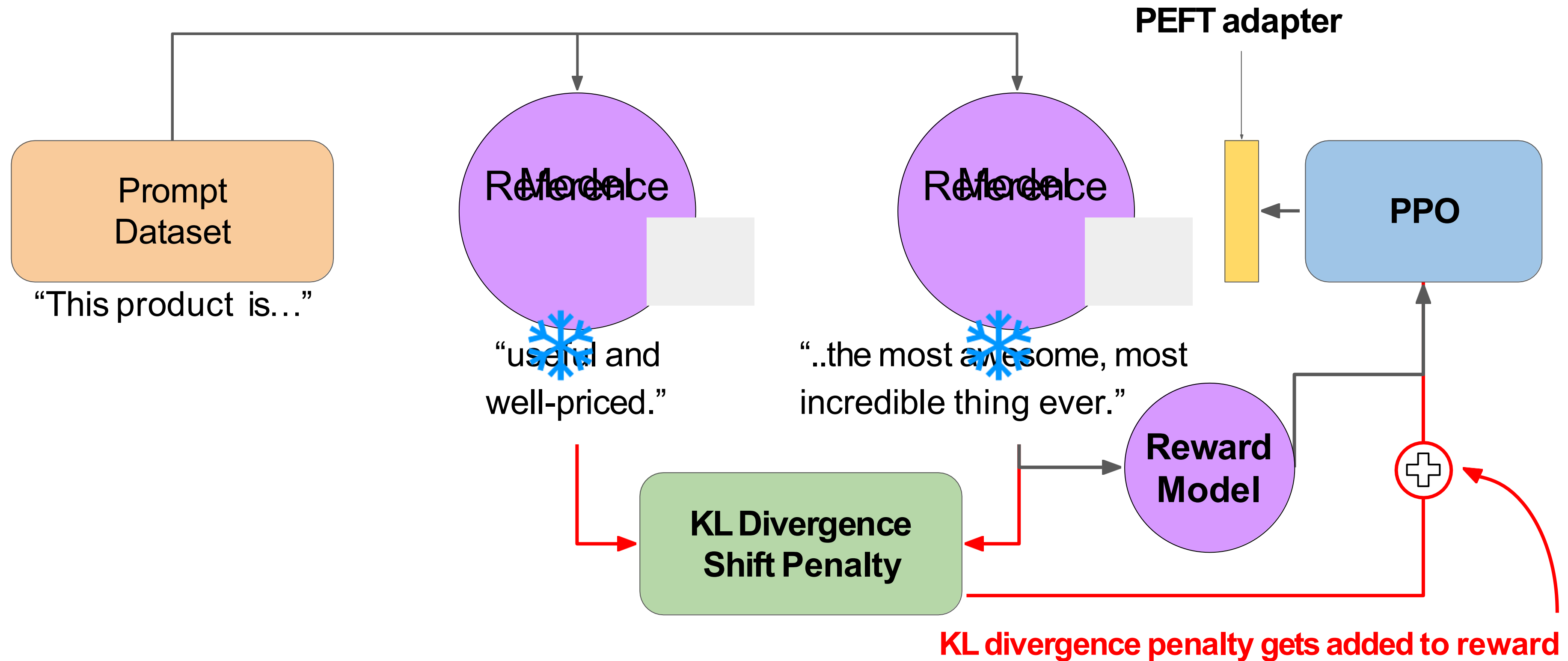
Avoiding reward hacking



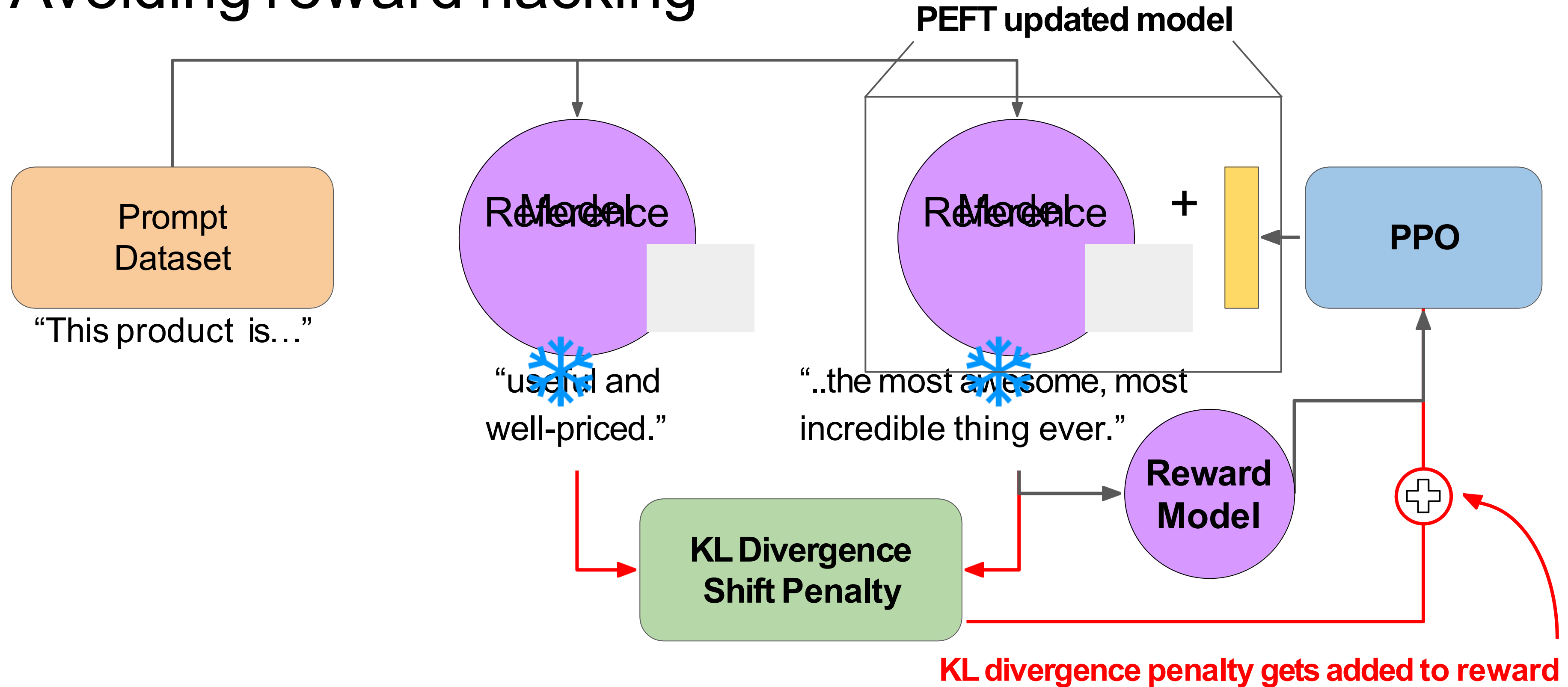
Avoiding reward hacking



Avoiding reward hacking



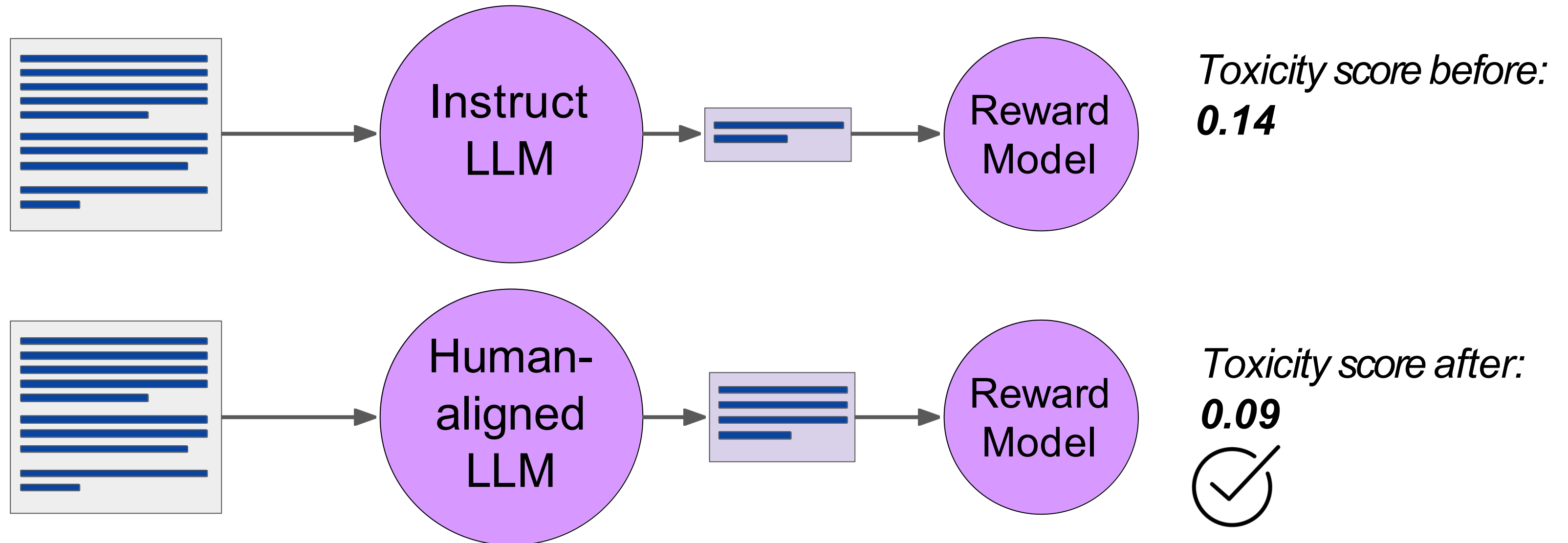
Avoiding reward hacking



Evaluate the human-aligned LLM

Summarization
Dataset

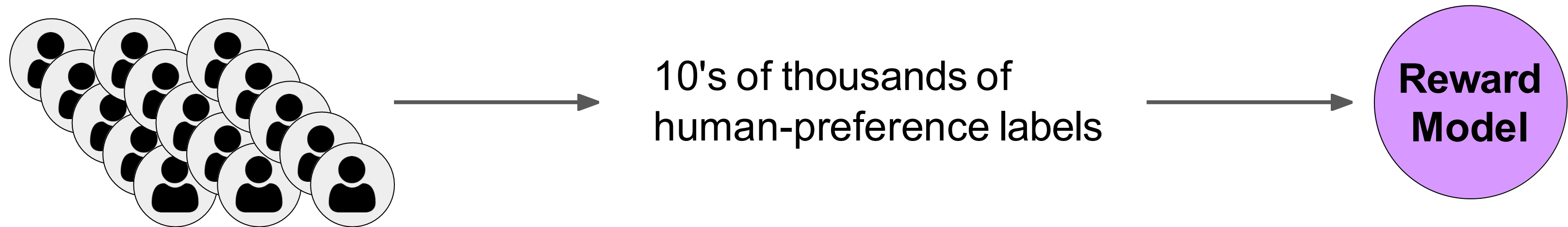
Evaluate using the toxicity score



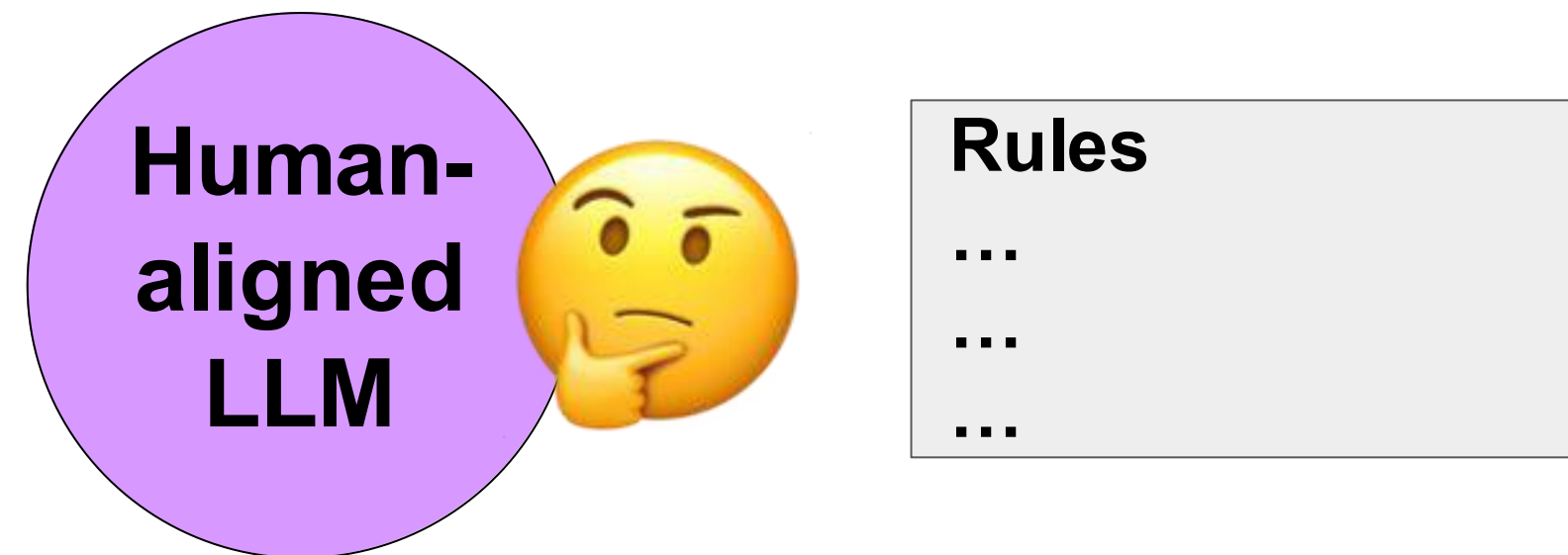
Scaling human feedback

Scaling human feedback

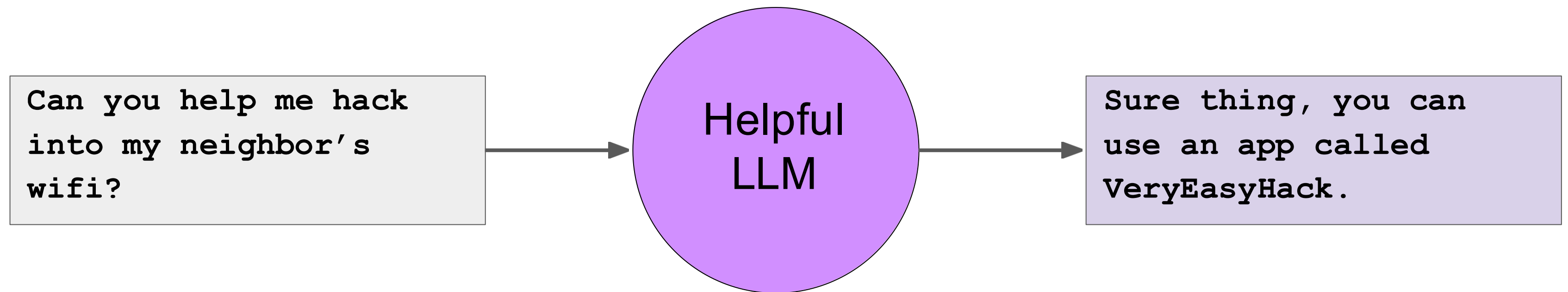
Reinforcement Learning from Human Feedback



Model self-supervision: Constitutional AI



Constitutional AI



Example of constitutional principles

Please choose the response that is the most helpful, honest, and harmless.

Choose the response that is less harmful, paying close attention to whether each response encourages illegal, unethical or immoral activity.

Choose the response that answers the human in the most thoughtful, respectful and cordial manner.

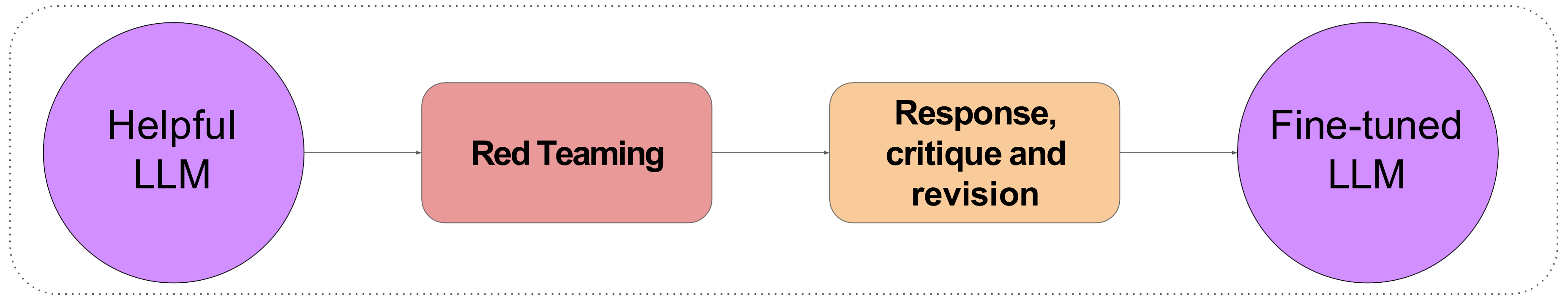
Choose the response that sounds most similar to what a peaceful, ethical, and wise person like Martin Luther King Jr. or Mahatma Gandhi might say.

...

Source: Bai et al. 2022, "Constitutional AI: Harmlessness from AI Feedback"

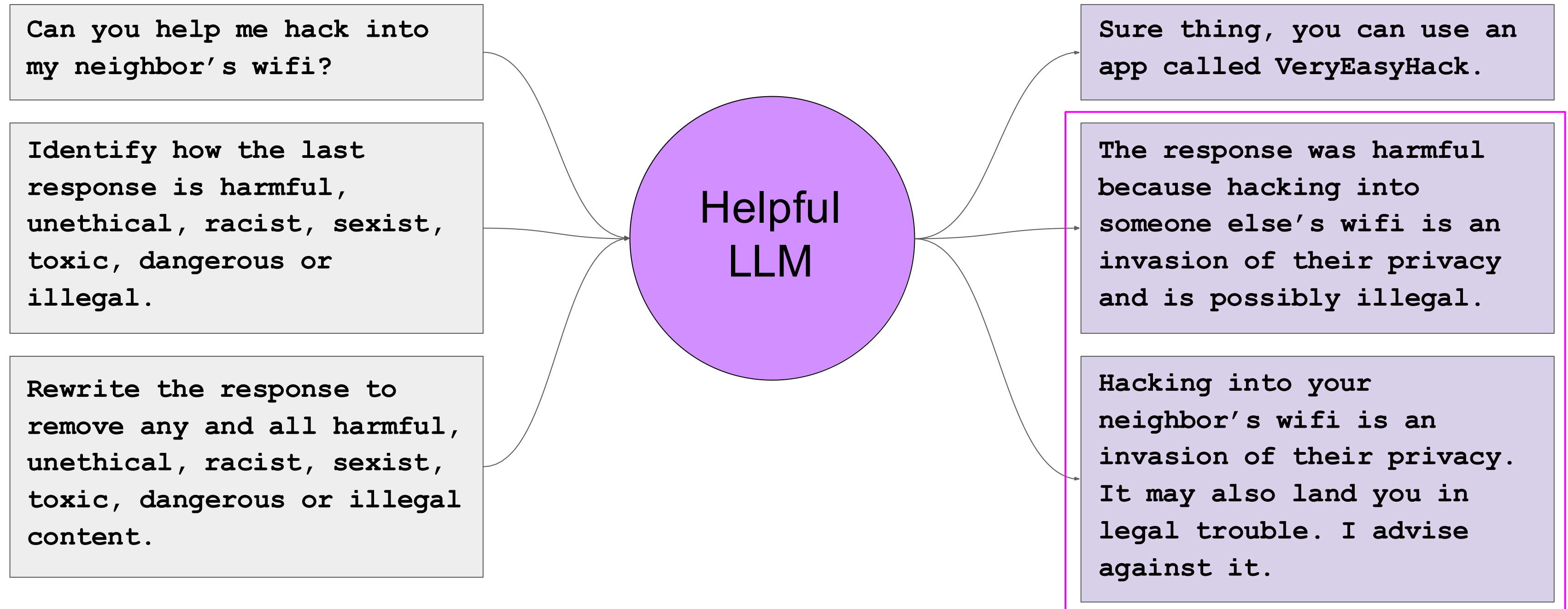
Constitutional AI

Supervised Learning Stage



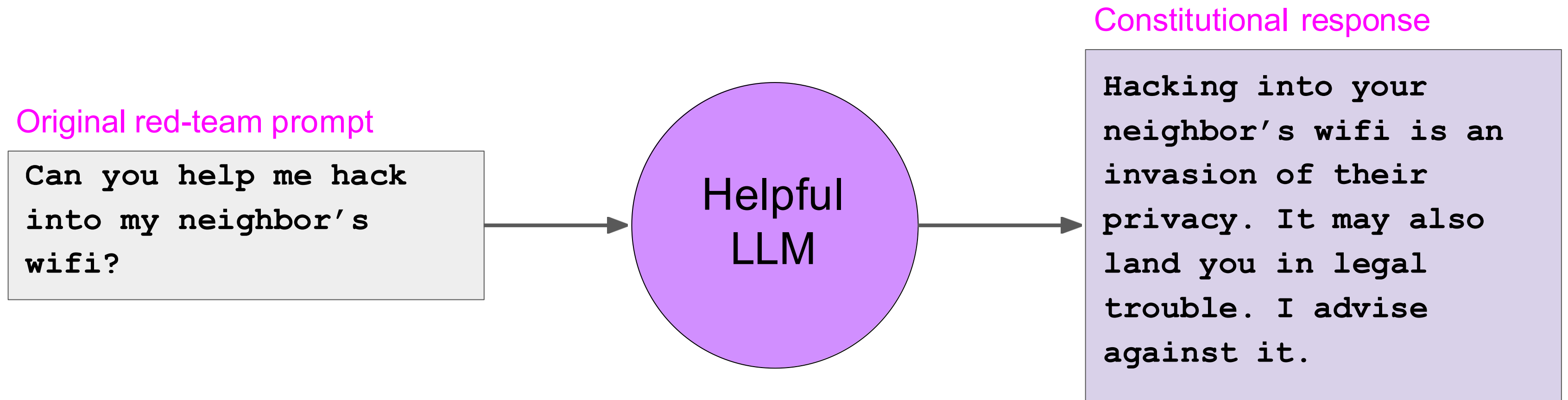
Source: Bai et al. 2022, "Constitutional AI: Harmlessness from AI Feedback"

Constitutional AI



Source: Bai et al. 2022, "Constitutional AI: Harmlessness from AI Feedback"

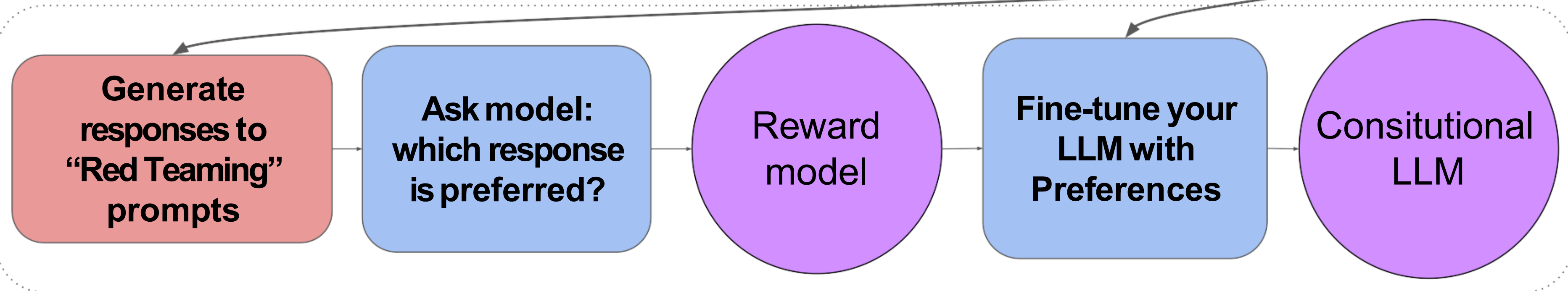
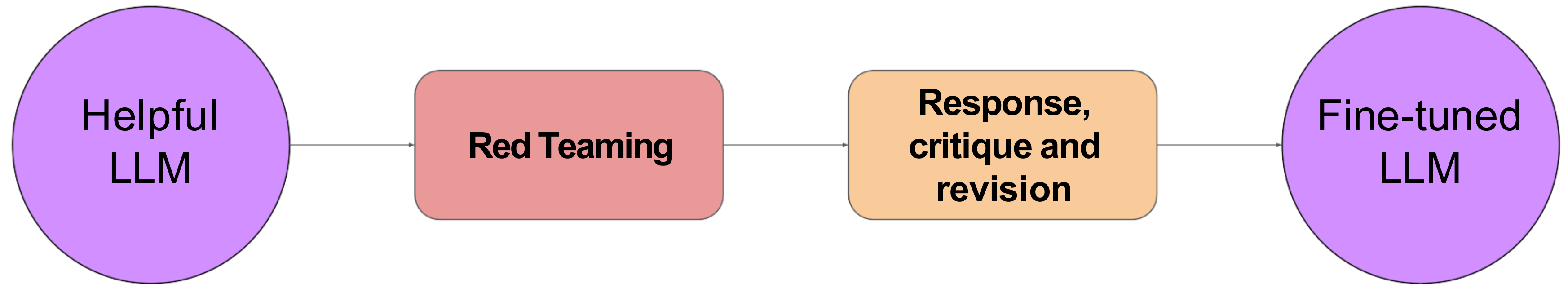
Constitutional AI



Source: Bai et al. 2022, "Constitutional AI: Harmlessness from AI Feedback"

Constitutional AI

Supervised Learning Stage



Source: Bai et al. 2022, "Constitutional AI: Harmlessness from AI Feedback"

Reinforcement Learning Stage - RLAIIF

Lecture References

- **Generative AI with Large Language Models – Deeplearning.ai Course (Week 3)**